Circular Economy as growth lever for Agrasar Bharat

Reimagining Water and Solid Waste Economy

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Foreword

India has demonstrated that it can build environmental infrastructure at scale. The challenge before us now is to transform that infrastructure into an integrated resource economy that secures water, recovers value from waste, and sustains growth in a resource-constrained future.

As India advances towards the vision of Viksit Bharat@2047, the country's growth trajectory will increasingly depend on how productively and sustainably it manages its natural resources. Rapid urbanisation, industrial expansion, climate variability, and changing consumption patterns are placing unprecedented pressures on water, waste, and environmental systems. At the same time, emerging growth sectors such as semiconductors, data centers, green hydrogen, advanced manufacturing, and Global Capability Centers are expected to significantly increase demand for water, energy, and raw materials.

The central proposition of this report is both simple: **water, wastewater, solid waste and recovered materials can no longer be planned and managed as separate sectors.** They are interconnected elements of a single resource system. The transition to a circular economy is therefore not merely an environmental imperative; it is an economic necessity for ensuring long-term resource security, industrial competitiveness, and urban resilience.

Three messages deserve particular emphasis. Firstly, India's environmental challenge is increasingly an institutional challenge, not an infrastructure challenge. Significant investments have already been made through flagship missions such as AMRUT, Jal Jeevan Mission, Swachh Bharat Mission, Smart Cities, and the National Mission for Clean Ganga. The next phase must shift the focus from creating assets to making them work – strengthening service delivery, operational performance, asset management, and accountability. Success should be measured not by infrastructure created, but by outcomes delivered: water security, resource recovery, service reliability, and environmental impact.

Secondly, resource circularity must become a core pillar of India's growth strategy. Wastewater reuse, waste-to-energy solutions, secondary materials markets, critical mineral recovery, and producer responsibility frameworks represent significant economic opportunities. However, value will be created only when markets are viable – through clear standards, assured demand, appropriate pricing and incentives, alongside the policy, regulatory reforms and institutional arrangements to support scale.

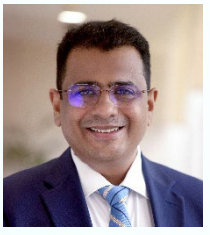
Finally, technology and data must evolve from monitoring to decision-making. Digital platforms, smart metering, SCADA systems, AI-enabled analytics, material traceability, and predictive asset management can substantially improve efficiency, transparency, and service outcomes. The objective should be to build decision infrastructure that strengthens planning, operations, and governance across the resource value chain.

Realising this vision will require a new enabling architecture. Policy and regulation must create the right incentives for resource recovery, reuse, and circular business models. Financing must blend public investment with private capital, outcome-based funding, and innovative risk-sharing instruments. Institutions will need stronger technical capacities, professionalized utilities, and integrated governance arrangements. Equally important will be the adoption of fit-for-purpose technologies supported by digital intelligence and real-time operational visibility.

Going forward, three priorities can help accelerate this transition:

1. **Institutionalise resource productivity as a core urban performance metric**, shifting the focus from infrastructure creation to outcomes delivered – water reuse, material recovery, emissions reduction, and service quality.
2. **Catalyse markets for treated wastewater and secondary materials** through quality standards, procurement mandates, certification mechanisms, and market platforms that create predictable demand for recovered resources.
3. **Promote integrated and regional resource planning** by encouraging circular economy cells, regional waste management clusters, digitally enabled utilities, and investment-ready project structures capable of attracting blended and private finance.

India's next decade of environmental infrastructure development will be defined not by how much more we builds, but by how effectively we conserve, recover, and reuse what we already have. The opportunity is to move from an infrastructure-led approach to an outcomes-led, digitally enabled, and circular model of resource management. This report offers a timely roadmap for that transition – one that can help India secure its resources, strengthen the resilience of its cities and industries, and create new avenues for sustainable growth. These are essential building blocks for an *Agrasar Bharat* – an India progressing with confidence towards the vision of Viksit Bharat@2047.



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Foreword

"Municipal waste need not remain a cost to be managed. With regional scale, reliable offtake and credible commercial structures, it can become a source of materials, energy, investment and employment"

India enters this decade with a rare advantage: the world's largest waste management movement, and a policy architecture that has begun to treat discarded material not as a disposal burden but as an economic input. For a decade, India has measured progress in tonnes of waste collected, processed and cities declared clean. These are commendable outcomes, and they have served the country well. But collection and processing are not the same as value, and the three need to be planned together rather than one after the other. The decade behind us established access. The decade ahead offers something considerably more valuable — the conversion of material flows into income, employment and industrial feedstock, in the same towns and districts where that material is generated.

Consider what is already changing, Waste that once sat in landfills purely as a disposal problem is beginning to be viewed as feedstock, for compressed biogas, for refuse-derived fuel, for compost, for energy recovered at scale. Smaller cities that could never justify a standalone processing facility are increasingly served by regional clusters instead, sharing infrastructure that no single municipality could finance alone. Markets for these outputs, and the institutions needed to sustain them, are increasingly the binding constraint, not the underlying technology.

The scale of that opportunity is now well documented. The Ministry of Environment, Forest and Climate Change has assessed that India's circular economy could represent a market opportunity of approximately USD 2 trillion and support around 10 million jobs by 2050. Circularity is further estimated to generate additional annual value of about USD 218 billion by 2030, rising to roughly USD 624 billion by 2050. These figures describe value that is presently being forgone.

This publication advances a fundamental proposition: solid waste management should be viewed not only as an operational challenge but also as a compelling economic opportunity. Rather than treating waste management as a municipal cost center focused on expenditure optimization, stakeholders should recognize its potential as a scalable and commercially viable sector. When supported by regional aggregation, predictable tipping-fee mechanisms, and dependable demand for recovered energy and recyclable materials, the sector can attract substantial private investment, reducing reliance on grants and viability gap funding.

Investors, developers, and policymakers are encouraged to evaluate the solid waste sector through this strategic lens. The sections that follow extend beyond a descriptive assessment of India's waste collection and processing infrastructure. More importantly, they provide a forward-looking perspective on where regional waste economies are likely to emerge, highlighting the locations and market conditions where value creation, investment opportunities, and sustainable returns are expected to materialize first.



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Foreword

"India's water future will be determined not only by the development of new sources and infrastructure, but by the institutions that protect every drop, distribute it efficiently and return it to productive use."

Two cities may have similar water infrastructure and yet deliver very different levels of service. In one, water reaches households reliably. In the other, a substantial share is lost before it reaches the consumer. The difference often lies not in the infrastructure itself, but in how it is managed.

This is an important theme running through the publication's account of India's water sector. From my experience, many of the sector's challenges are no longer confined to source availability, treatment capacity or network design. Increasingly, they concern the systems required to operate infrastructure effectively, including metering, billing, tariff design, maintenance and institutional accountability. As India continues to expand water infrastructure, greater attention must now be given to these operational foundations.

There is considerable opportunity in making this shift. A treatment plant should not be viewed solely as a facility for meeting service standards. With appropriate pricing, reliable offtake arrangements and accountable operators, it can also support the recovery of water, energy and nutrients. Similarly, reducing losses from unbilled and unaccounted-for water can provide cities with additional supply at a lower cost than developing entirely new sources. In many cases, the infrastructure already exists. The priority is to improve how it is operated and governed.

Energy consumption in water infrastructure and service delivery is another overlooked variable which accounts for 40 to 60 per cent of the cost of urban water management, and yet it is usually treated as an operating expense to be absorbed after commissioning rather than a design parameter to be optimised before it. Efficiency improvements of the order this report describes — savings around INR 3,200 crore a year and roughly 600 MW of avoided demand — are available largely through decisions taken at the drawing board.

My own conclusion is that technology selection must be judged against the operating reality of the agency that will run it — its staffing, its financing, its maintenance regime — and not against the specification sheet. The most appropriate solution is the one that still works in year fifteen. This will become increasingly important as India's urban and industrial water requirements grow. The country's water security over the coming decade will depend not only on identifying new sources, but also on protecting, distributing and reusing the water already available. I hope this publication contributes to that effort and supports those working to build more reliable, efficient and sustainable water systems.



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Foreword

"Circularity is not a trend to be embraced when convenient, it is a boundary defined by the planet's finite resources, requiring economies and societies to rethink how value is created, consumed, and regenerated."

Every economy that has ever grown has done so by taking more from the earth than it returned. For much of human history, the planet's capacity to absorb that imbalance was large enough that the consequences were not always visible. Today, however, that capacity is under growing pressure, and continuing to assume that natural systems can absorb increasing levels of extraction and waste only delays the action that is needed.

India's growth journey is both necessary and important. As the country continues to expand infrastructure, improve public services, and raise living standards for millions of people, the demand for resources will naturally increase. The more important question is not whether growth should continue, but whether the products, materials, and resources that support this growth are designed to remain in use for as long as possible rather than being discarded after a single lifecycle.

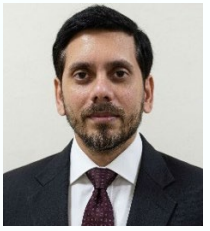
This is why circularity deserves to be treated as a necessity rather than a preference. Unlike trends that may change with time, the limits of our natural resources remain constant. As populations grow and economies expand, it becomes increasingly important to find ways of creating more value from the resources we already have. Every tonne of material recovered and returned to productive use reduces the need for fresh extraction, while every product designed for reuse, repair, remanufacturing, or recycling helps build a more resilient and resource-efficient economy.

The circular economy is not a separate chapter appended to India's growth story. It is the design principle by which that story continues to be told without exhausting the conditions that make it possible. This publication treats the circular economy as exactly that the mechanism by which materials, once used, are not lost but returned, so that the country's development does not outpace the planet's capacity to sustain it.

When waste is discussed, the conversation often centres on municipal solid waste. Yet some of the most significant opportunities for circularity lie beyond it. Textile waste, plastic waste, agricultural waste, construction and demolition waste, electronic waste, and industrial waste together represent enormous reservoirs of materials that are frequently discarded despite retaining substantial economic value. The next wave of the circular economy will be defined not only by how effectively cities manage municipal waste, but also by how successfully industries, businesses, farmers, and governments transform these broader waste streams into resources that can be reused, recycled, and reintegrated into the economy. Unlocking this value will be critical to improving resource efficiency, reducing dependence on virgin materials, and supporting long-term sustainable growth.

This thought leadership document examines water, solid waste, resource recovery and the circular economy not as separate disciplines, but as different expressions of the same underlying choice. For decades, growth has been built on a straight line, Make, use, discard, and begin again with new resources. That line was only ever available to us because the planet was willing to keep absorbing what we left behind. Mother Earth is not willing to keep doing so indefinitely.

"The world must stop practicing linearity and start practicing circularity now, otherwise we won't leave a safe and sustainable planet for our future generation"



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List of abbreviations

Abbreviation	Full form
AI	Artificial Intelligence
AMI	Advanced Metering Infrastructure
AMRUT	Atal Mission for Rejuvenation and Urban Transformation
BCC	Behaviour Change Communication
BCM	Billion Cubic Metres
Bio-CNG	Bio-Compressed Natural Gas
C&D	Construction and Demolition
CA	Civil Appeal
CAG	Comptroller and Auditor General
CBG	Compressed Biogas
CIPET	Central Institute of Petrochemicals Engineering and Technology
CO_2e	Carbon Dioxide Equivalent
CPCB	Central Pollution Control Board
CSE	Centre for Science and Environment
CSR	Corporate Social Responsibility
DMA	District Metered Area
DMF	District Mineral Foundation
DPR	Detailed Project Report
DRAP	Dumpsite Remediation Accelerator Programme
DRIP	Dam Rehabilitation and Improvement Project
DSS	Decision Support System
EBWGR	Extended Bulk Waste Generator Responsibility
EPR	Extended Producer Responsibility
FHTCs	Functional Household Tap Connections
FSSAI	Food Safety and Standards Authority of India
FSSM	Faecal Sludge and Septage Management
FY	Financial Year
GCC	Global Capability Centres
GDP	Gross Domestic Product
GeM	Government e-Marketplace
GHG	Greenhouse Gas
GIS	Geographic Information System
GOBARdhan	Galvanizing Organic Bio-Agro Resources Dhan
GP	Gram Panchayat
HAM	Hybrid Annuity Model
HUL	Hindustan Unilever Limited
IEC	Information, Education and Communication
IoT	Internet of Things
IS	Indian Standard
JJM	Jal Jeevan Mission
KPI	Key Performance Indicator

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List of abbreviations

Abbreviation	Full form
kt	Kilotonnes
LIB	Lithium-Ion Battery
MIS	Management Information System
ML	Machine Learning
MLD	Million Litres per Day
MoEFCC	Ministry of Environment, Forest and Climate Change
MRF	Material Recovery Facility
MSW	Municipal Solid Waste
MT	Metric Tonne
MW	Megawatt
MWe	Megawatt Electrical
NAMASTE	National Action for Mechanised Sanitation Ecosystem
NRW	Non-Revenue Water
PIBO	Producer, Importer and Brand Owner
PMU	Project Management Unit
PPE	Personal Protective Equipment
PPP	Public Private Partnership
PV	Photovoltaic
PWD	Public Works Department
PWM	Plastic Waste Management
QR	Quick Response
RDF	Refuse Derived Fuel
SBM	Swachh Bharat Mission
SBM-U	Swachh Bharat Mission Urban
SC	Supreme Court
SCADA	Supervisory Control and Data Acquisition
SHG	Self-Help Group
SLAs	Service Level Agreements
SPV	Special Purpose Vehicle
STP	Sewage Treatment Plant
SWaCH	Solid Waste Collection and Handling
SWM	Solid Waste Management
TSU	Technical Support Unit
UCF	Urban Challenge Fund
ULB	Urban Local Body
UNDP	United Nations Development Programme
UT	Union Territory
VGF	Viability Gap Funding
VWSC	Village Water and Sanitation Committee
WtE	Waste-to-Energy
WTP	Water Treatment Plant

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Section 1:

India's environmental infrastructure: From access to resource security



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Environmental infrastructure development in India is entering a new phase. Policy priorities are shifting beyond asset creation towards resource security, service reliability and long-term sustainability. Water, sanitation, and waste management are increasingly recognised as interconnected systems that influence public health, economic growth, and climate resilience.

'This evolution reflects a broader global understanding that infrastructure alone does not create security. The real objective is to ensure that resources remain available, services remain reliable, and systems remain resilient under growing environmental and economic pressures.'

India's flagship missions increasingly reflect this systems-based approach. Jal Jeevan Mission aims to link household water access with source sustainability, recharge, rainwater harvesting and greywater management. AMRUT 2.0 integrates water supply, sewerage, septage management, treated-water reuse and waterbody rejuvenation within a common planning framework. Swachh Bharat Mission looks beyond sanitation access to include wastewater treatment, resource recovery, and solid waste management.

Together, the overall national agenda shows a shift from managing isolated programmes towards managing interconnected resource flows across the entire water and sanitation value chain.

The defining challenge for the next decade is not how much infrastructure India can build, but how effectively existing and future infrastructure performs. Research increasingly shows that long-term outcomes depend on resource productivity, efficiency, and recovery rather than the quantity of assets created. Future success could therefore be measured by water secured, resources reused, value recovered, and services delivered reliably and affordably. Investments that improve system performance, strengthen institutions and enable circular resource flows is expected to generate greater resilience than those focused solely on expanding physical infrastructure.

1.1 The new resource equation: Growth, scarcity, and circularity

Resource stress is increasingly a consequence of economic success. Growth tends to concentrate around established economic centres, regardless of local resource availability. As demand accumulates over decades, many of the country's most productive regions find themselves operating beyond the limits of their natural resource base. Future resilience is expected to depend less on finding new resources and more on improving efficiency, reuse and resource circularity.

Several structural shifts are redefining the future of water, sanitation and waste management in India.



Urbanisation and economic transformation are concentrating and expanding resource demand in regions already exposed to water stress and climate risks



Climate uncertainty is reducing the reliability of traditional infrastructure planning and increasing the need for resilience across systems



Resource constraints are shifting the focus from availability alone to the efficient allocation, movement, treatment and reuse of water



Waste valorisation is creating new opportunities to recover water, energy, nutrients and materials from waste streams.

The circular economy agenda is reinforcing this shift. Extended Producer Responsibility (EPR) frameworks for plastics, e-waste, batteries, tyres and used oil are creating new incentives for collection, recycling and resource recovery. As a result, environmental infrastructure is no longer confined to service delivery. It is becoming part of a broader ecosystem that combines recovery infrastructure, digital traceability, regulatory compliance and secondary-material markets to unlock value from waste streams.

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The question is no longer whether infrastructure can solve resource constraints. Global experience shows that it can, when supported by effective institutions, resource management and long-term planning. For India, the next stage of reform is therefore not a choice between infrastructure and sustainability, but the integration of both. As water stress and climate risks grow, future resilience could depend as much on efficiency, reuse and governance as on the creation of new assets.

1.2 The debates shaping the next decade

The next phase of India's environmental infrastructure transition might be defined less by engineering choices and more by interdisciplinary decisions on efficiency, governance, finance and resource management.

Energy efficiency as a water-security strategy
The focus is shifting from supplying energy to improving energy productivity. Efficient pumping, treatment and wastewater operations can reduce costs, improve performance and strengthen long-term sustainability.

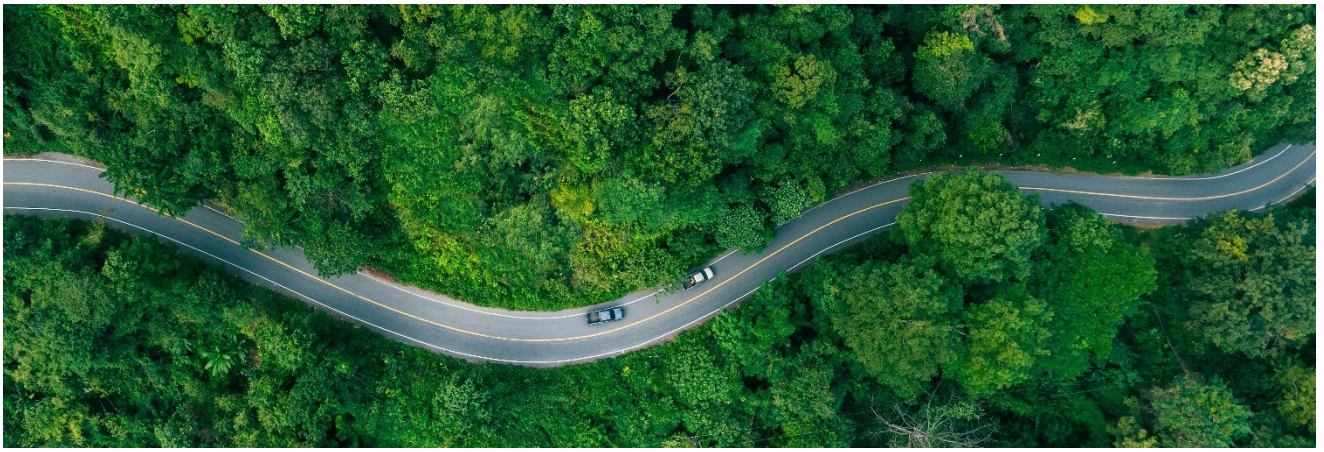
From water availability to water security
Water security is about more than water availability. It depends on reliable, affordable and resilient services. The challenge is shifting from managing shortages to strengthening source protection, efficiency, demand management and reuse.

Centralised or decentralised? The misplaced question
The choice is not between centralised and decentralised infrastructure. The priority is selecting fit-for-purpose solutions that match local conditions, resource availability, service needs and lifecycle costs. Successful infrastructure planning is expected to increasingly depend on flexibility rather than standardisation.

From asset creation to asset performance
India has made significant progress in expanding water and sanitation infrastructure. The next challenge is ensuring that these assets consistently deliver outcomes. Functionality, operational efficiency, financial sustainability, and service quality are becoming more important than construction targets. Success could be measured by performance, not by assets alone.

Rethinking infrastructure finance
Public finance is expected to remain essential for universal water and sanitation services. However, growing investment needs are creating opportunities for blended finance and performance-based partnerships. The key challenge is not choosing between public and private capital, but matching financing structures to project risks, revenue potential and development objectives.

From universal coverage to equitable service
Access alone does not guarantee inclusion. Differences in reliability, quality, affordability and service levels often persist across communities. The next generation of policy must focus not only on who is connected, but also on who receives safe, affordable and reliable services. Equity could increasingly become a measure of infrastructure performance.



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Digitalisation and circularity: From technology to outcomes

Digital systems create value when they improve decisions, operations and accountability. Similarly, circularity creates value only when recovered resources are productively reused. The one of the greatest opportunity lies in combining digital intelligence with circular infrastructure to reduce losses, improve performance and maximise resource recovery.

'At the same time, human judgement, experience, and priorities cannot be replaced by or outsourced to machines. Authorities/decision-making systems need to be increasingly aware and more sensitive towards this characteristic binding constraint of deploying artificial intelligence and other forms of automation in systems.'

India's environmental infrastructure agenda is moving from coverage to service continuity, from disposal to resource recovery, and from isolated assets to integrated systems. Across water, sanitation and waste management, success increasingly depends on reliable data, capable institutions, appropriate technology, sustainable finance, skilled operations and measurable outcomes. The following sections examine these transitions through the lenses of water security, solid waste management, EPR and circular-economy opportunity.

India's environmental infrastructure transition could place equity alongside efficiency and sustainability. As resource pressures grow, the allocation of public resources cannot be determined solely by purchasing power or geography. The objective should be reliable, affordable and equitable service for all. Achieving this requires more than infrastructure investment. It requires transparent and accountable institutions, effective regulation and safeguards against regulatory capture. It also requires meaningful stakeholder participation supported by strong institutional and community capacity. The long-term success of environmental infrastructure could depend not only on the services delivered, but on how fairly their benefits are shared.



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Section 2:

From source to circularity: securing India's water value chain

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2.1 Reimagining water in India's economic future

This section presents water as a single value chain that underpins India's economic growth. Real progress requires moving beyond fragmented programmes toward integrated planning across the entire water cycle. That shift is already underway.

India envisions its water resources to underwrite an economy that wants to grow nearly eightfold by 2050 - while the resource base itself shrinks, and the country holds nearly 18 per cent of the world's population on roughly 4 per cent of its freshwater[1]. This imbalance is the reason water needs to be treated as a 'stakeholder,' a part of the value chain, not as a peripheral utility.

Upstream, initiatives such as Atal Bhujal Yojana, Jal Shakti Abhiyan: Catch the Rain, Pradhan Mantri Krishi Sinchayee Yojana watershed development and Namami Gange Programme focus on source protection and resource replenishment. Midstream, programmes such as the National Hydrology Project, DRIP and AMRUT strengthen water storage, monitoring, treatment and distribution. Downstream, Jal Jeevan Mission, AMRUT 2.0 and Swachh Bharat Mission deliver water and sanitation services while increasing emphasis on wastewater treatment, septage management and reuse.

Together, these interventions form a continuous chain from source protection to service delivery and resource recovery. Viewed through this lens, India is steadily building the foundations of an integrated water security system. Policy is following the same logic, shifting from an 'asset creation' mindset toward sustained service delivery - a shift made more urgent by the Viksit Bharat 2047 vision, whose push into water-intensive sectors like semiconductors, data centres, and global capability centres (GCCs) could triple current water demand by 2030. Water security, in other words, is a prerequisite for that growth, not a compliance box to tick alongside the same.

2.2 Evolution of India's water security landscape

The stress on India's water system is compounding from two directions at once. On the demand side, a rising population is tightening the per-capita calculation with the per-capita water availability set to fall to roughly 1,367 cubic metres by 2031, about 20 per cent below the international stress threshold of 1,700 cubic metres.[2] On the supply side, climate change is making that shrinking base less predictable. India's average temperature is projected to rise 1.2–1.8°C by 2050[3], intensifying evaporation and drought risk, a reality visible on ground. At the same time, the country recorded its ninth-driest January on record in the recent year of 2024, with north-west India near its lowest rainfall since 1901, even as the South Peninsula logged a 133 per cent surplus over normal in the same period.[4]

Rising consumption patterns, rising demand for a stressed resource

Demand itself is not just rising, it is reshaping around new, harder-to-serve areas. Manufacturing is one of the fastest-growing use cases by far, with more than tripling from 16 to 50 billion cubic metres (BCM) as semiconductors, data centres, and green hydrogen production all scale up. That trajectory is tied directly to India's ambition to grow from a USD4 trillion to a USD30–35 trillion economy by 2050, with manufacturing alone expected to contribute at least a quarter of GDP.[5]

1. Water: Driving Jobs and Prosperity, World Bank Group, March 2026
2. Per Capita Water Availability, Press Information Bureau, December 2022
3. G20 Climate Risk Atlas
4. India 2024: An Assessment of Extreme Weather Events, Down to Earth
5. India Achieves ₹14 Lakh Crore in Production, ₹5.3 Lakh Crore in Exports, Says DPIIT Secretary at CII Manufacturing Summit, Ministry of Commerce & Industry, Press Information Bureau, March 202

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Agriculture, however, remains a base throughout the entire period. Total water withdrawals are projected to rise from 949 BCM in 2010 to 1,058 BCM by 2050, with agriculture's own share of that total climbing from 77 to 81 per cent - tightening competition for reallocation even as the wider economy diversifies away from farming.[6]

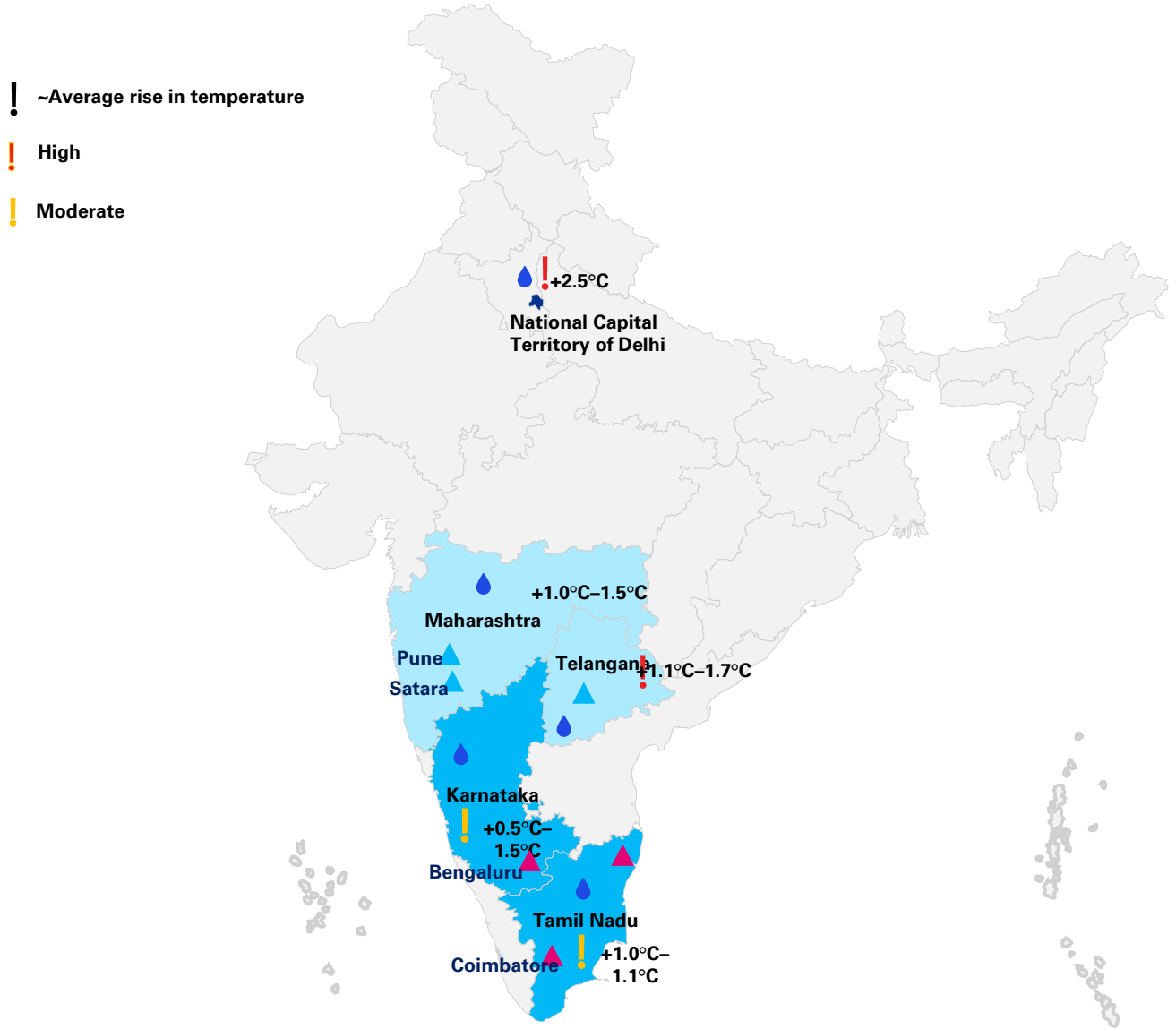
The overlap is geographical, as much as sectoral. India's leading data centre and GCC hubs, are largely concentrated in states such as Karnataka, Maharashtra, Tamil Nadu and Telangana with states such as Andhra Pradesh and Delhi NCR, also entering the fray. They are projected to witness 1-2°C rise in average temperatures[7], and many are being set up in districts already classified as having semi-critical or critical groundwater blocks alongside deficient rainfall. In short, the one of the fastest-growing pockets of water demand are landing precisely where supply is already tight.

The persistence of many water-sector challenges is not primarily a consequence of technical incapacity. In many cases, the required technologies, finance, and knowledge already exist. Their persistence reflects institutional arrangements in which the costs of degradation are socialised while the benefits of unsustainable behaviour remain concentrated. Lasting reform therefore requires not better engineering alone, but a deliberate reshaping of incentives, accountability, and stakeholder interests so that more actors benefit from solving the problem than from perpetuating it.



6. Reallocating Water for India's Growth, CEEW, December 2020
7. G20 Climate Risk Atlas

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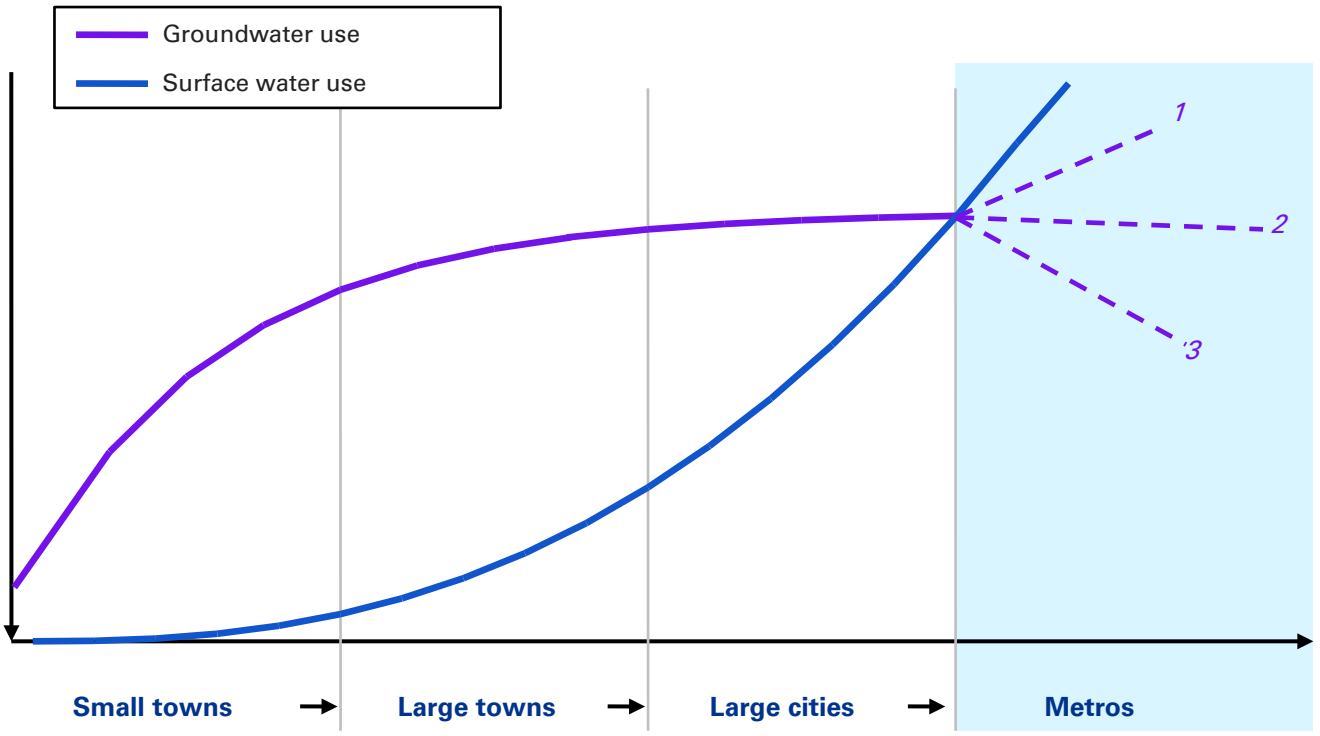
Sources: State Action Plans on Climate Change, India Meteorological Department (accessed on 3 July 2026)*, National Compilation on Dynamic Ground Water Resources of India 2025*

Shift from groundwater sources to surface water sources

Over the past decade, a significant shift has been observed from groundwater-based water supply systems to surface water sources to ensure long-term sustainability and water security. Under national initiatives such as AMRUT, AMRUT 2.0, and the Jal Jeevan Mission (JJM), greater emphasis has been placed on utilizing surface water as the primary source of supply, with groundwater being considered only where surface water sources are unavailable. In rural areas, nearly 50% of water supply schemes are now planned around surface water sources, compared to approximately 15% in the past. Similarly, several urban centers, including Chandigarh, Amritsar, Delhi, Gurugram, and Nagpur, have transitioned from groundwater-dependent systems to predominantly surface water-based municipal water supply networks.

As settlements evolve from small towns to larger cities and eventually metropolitan regions, their primary water source gradually shifts from groundwater to surface water. Increasing population density, rising water demand, and the limitations of local aquifers make groundwater sources inadequate and often unsustainable. Hence, larger urban centers become increasingly dependent

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Small towns	Large towns	Large cities	Metros
Surface water use limited to ponds/water bodies Groundwater use significant, largely through wells; sometimes through springs	Surface water use may increase as formal water supplies established Groundwater usage shifts outwards into suburbs and peri-urban areas; groundwater and surface water are transported from peri-urban/rural areas inward	Near equal share of surface water and groundwater, although much of the groundwater share 'unacknowledged' in civic water supply; Large water transports through tanker-supply; Wells are drilled into aquifers feeding springs	Large share of 'imported' formal surface water supply, with even greater share of 'unacknowledged' groundwater abstraction and transport in suburbs and peri-urban areas; uncertainty in the projected trends of groundwater use (1, 2 or 3) - possibly different in different metros.

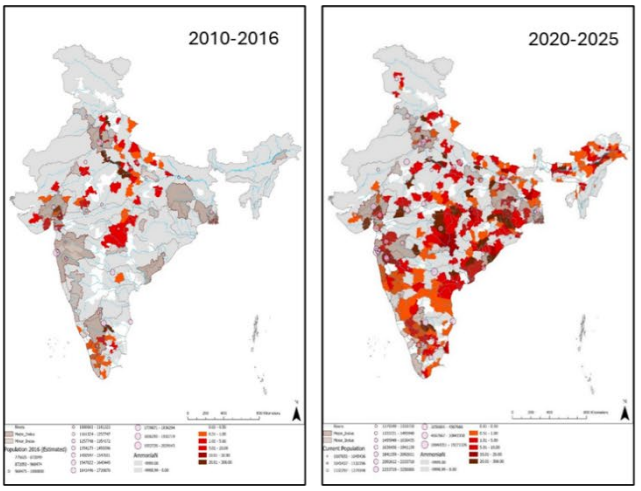
Sources: Urban Water Systems in India: A Way Forward, Indian Council for Research on International Economic Relations (ICRIER), Mihir Shah 2014



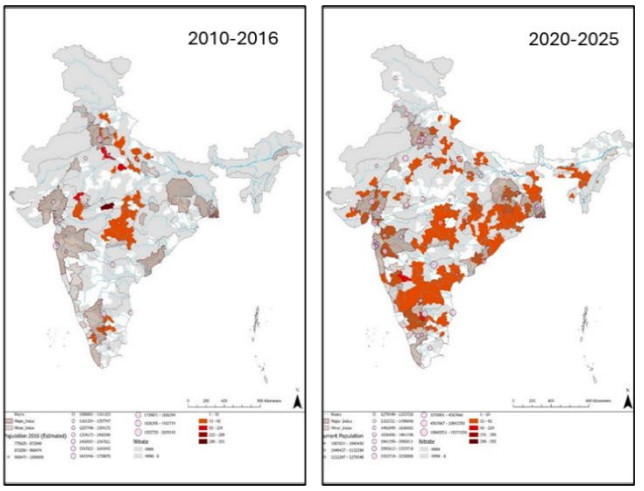
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Rising Water Contamination due to lack of wastewater collection and treatment

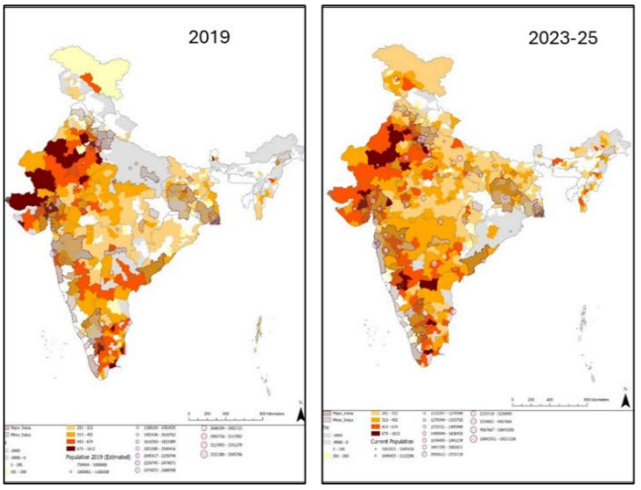
India's water scarcity challenge is no longer limited to the availability of water resources; it is increasingly becoming a question of water quality. Rapid urbanization, industrial growth, discharge of untreated sewage, agricultural runoff, and contamination of surface and groundwater sources have significantly reduced the volume of water that is fit for human consumption, industrial use, and ecosystem sustainability. As pollution levels rise, a substantial portion of available freshwater resources become unusable without treatment, consequently exacerbating the country's water stress. In many regions, water bodies are impacted by high levels of organic pollutants, nutrients, pathogens, heavy metals, emerging contaminants, and salinity, posing serious risks to public health and economic development.



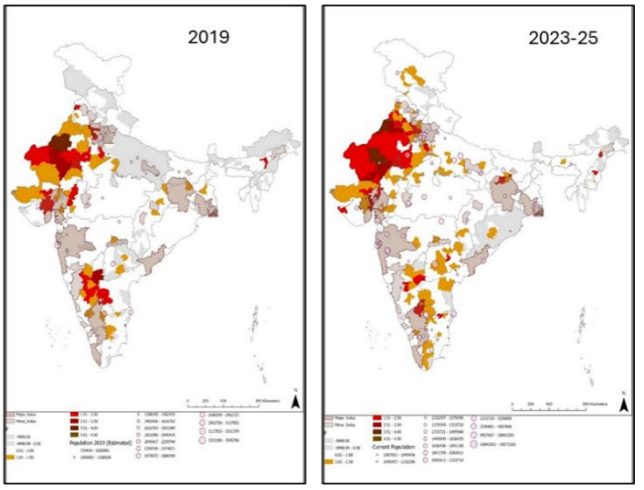
Total concentration of Ammonia-N (mg N/L) in surface water



Total concentration of Nitrate-N (mg N/L) in surface water



Total concentration of Total Hardness (mg/L as $CaCO_3$) in groundwater



Total concentration of Fluoride – F (mg/L) in groundwater

Sources: National Water Data Portal, Central Ground Water Board reports and Central Pollution Control Board reports, 2010-2025

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Typically, when water is sourced from surface water bodies such as rivers, lakes, or canals, a conventional Water Treatment Plant (WTP) is installed to remove pathogens and other contaminants. The cost of constructing a conventional WTP in India generally ranges from INR 30 lakh/MLD to INR 60 lakh/MLD[8], depending on factors such as plant capacity, location, and raw water quality. However, deteriorating source water quality and increasing land constraints in urban areas have created a growing need for advanced water treatment infrastructure. The cost of advanced water treatment plants varies considerably based on treatment capacity, water quality requirements, technology configuration, and project location. Currently, approximately 10% of upcoming WTP installations by capacity incorporate advanced treatment technologies; however, this share is expected to increase over the coming years if water sources contamination challenges are not addressed effectively.[9]

Rapid industrial growth is placing mounting pressure on freshwater resources, with a growing proportion of available water being diverted to industrial applications. This trend is expected to intensify with the expansion of water-intensive industries such as semiconductors, green hydrogen, and solar manufacturing, which require substantial quantities of high-quality water. Based on our analysis, investment costs for ultra-pure water treatment infrastructure can range between INR 40-200 crore per MLD, while process-grade water treatment facilities typically require investments of INR 10-50 crore per MLD.[10] The final capital expenditure is influenced by factors such as industry-specific water quality requirements, plant scale, source water characteristics, and site-specific considerations.

Investment trends

India has been very proactive in its policy response, and one of the clearest indications come with the large-scale national mission outlays in water and sanitation. For instance, the Jal Jeevan Mission's budgetary allocation rose from INR 50,011 crores in FY2021-22[11] to INR 70,000 crores in FY2023-24[12], and with its original INR 3.60 lakh crore outlay nearly fully spent; the mission has been extended to 2028 under JJM 2.0 with an enhanced INR 8.9 lakh crore outlay.[13]

AMRUT has followed the same arc, scaling from an approved INR 77,640 crores[14] under AMRUT 1.0 to a total indicative outlay of INR 2,77,000 crores[15] under AMRUT 2.0. This pattern - increasing budgets and longer timelines repeats across Namami Gange and Atal Bhujal Yojana. Together, it tells a consistent story: water security has become a sustained investment priority rather than a one-time infrastructure push, and spending is visibly moving from access creation toward reliability, reuse, and basin restoration.



8. KPMG in India's 2025-26 Analysis
9. KPMG in India's 2025-26 Analysis
10. KPMG in India's 2025-26 Analysis
11. Government releases Rs. 5,968 Crore Central grant to 15 States under Jal Jeevan Mission for financial year 2021-22, Ministry of Jal Shakti, Press Information Bureau, May 2024
12. IN 2023-24, Gross Budgetary Support of Rs.70,000 Crore Proposed for Implementation of Jal Jeevan Mission, Ministry of Jal Shakti, Press Information Bureau, March 2023
13. Cabinet approves extension of Jal Jeevan Mission (JJM) period up to December 2028 with enhanced outlay and restructured implementation focusing on structural reforms in rural drinking water supply sector under JJM 2.0, PMIndia, 2026
14. AMRUT & AMRUT 2.0 ACHIEVEMENT, Press Information Bureau
15. Atal Mission for Rejuvenation and Urban Transformation -AMRUT 2.0, Ministry of Housing and Urban Affairs

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2.3 Hindrances to water security

Four structural gaps stand between India's investment and the outcomes it is meant to achieve: the resource itself, how efficiently the system runs, how well it can see its own losses, and how much is circulated back into the system.

Source sustainability

There are instances, where investment is robust, however, the resource is under direct strain. Nearly 25 per cent of India's groundwater assessment units are already over-exploited, semi-critical or critical, and National extraction now runs at about 61 per cent of assessed recharge.[16] That national figure hides a sharper local reality, though: 11 of India's 15 major river basins are already water-stressed, and roughly 75 per cent of the country's annual rainfall arrives in a four-month single monsoon window, rivers across peninsular and western India run strong for a season and dry for the rest of the year.[17]

Asset visibility and Non-Revenue Water (NRW)

A utility cannot manage what it cannot measure. India's water metering gap makes that literal. For instance, in Delhi, 66 per cent of required flow meters at primary reservoirs were installed, and 73 per cent of those were functional[22]. Delhi's transmission loss rose from ~16 per cent in 2017-18 to ~20 percent in 2021-22, and NRW ran at 51–53 per cent of daily supply, eroding an estimated INR 4,988 crore in revenue.[23] Digitalisation is underway - AMRUT has added SCADA systems to 230 water and 146 sewerage projects[24], with hundreds more approved under AMRUT 2.0 - but it is still being added project by project rather than embedded system-wide, and JJM's SCADA and IoT rollout for rural water schemes remains in the nascent stage.

Operational resilience and energy efficiency

The system also has limited resilience to the droughts and other extreme weather events that are becoming more prevalent with climate change, leaving around 600 million people vulnerable to severe water scarcity.[18] Much of the cost of keeping this system running is not visible in water bills but in electricity bills. Most of India's 23 million minor irrigation schemes depend on groundwater pumping[19], and energy typically makes up 40-60 per cent of the cost of urban water supply.[20] This also creates a major opportunity: improving energy efficiency could reduce water-supply energy costs by 25-40 per cent, saving an estimated INR 3,200 crore annually and cutting power demand by about 600 MW.[21]

Limited water circularity

The same gap is visible at the other end of the water cycle. Urban India generates 72,368 MLD of sewage but treats only 20,235 MLD, meaning that nearly 52,133 MLD (around 72 per cent) is still discharged untreated into rivers, lakes, or land. Even where treatment infrastructure exists, it is not always fully utilised. Operational treatment capacity stands at 26,869 MLD, compared with an installed capacity of 31,841 MLD.[25] The challenge is not only treatment but also reuse.

The binding constraints on reuse are commercial and institutional. Freshwater is cheaper at the point of use because its network is already built and its tariff subsidised, while recycled water carries the full cost of a distribution system that varies by region and no single agency clearly owns or operates.

16. India's hidden water subsidy is collapsing, DownToEarth, May 2026
17. Monsoon 2025: A Report, India Meteorological Department, 2025
18. Toward Climate-resilient River Systems in Chennai, February 2026
19. Ministry of Jal Shakti Releases the 6th Census Report On Minor Irrigation (MI) Schemes
20. Financing Bundled Municipal Energy Efficiency (Ee) Projects Through Energy Performance Contracts By Energy Service Companies, Alliance to Energy
21. Assistance to cities to improve energy use efficiency to tap Rs.6,000 cr saving potential per year
    Ministry of UD and EESL sign MoU to promote Energy Efficiency Programmes at no cost to cities
    Shri Venkaiah Naidu, Shri Piyush Goyal laud the joint initiative, Press Information Bureau, September, 2016
22. CAG (2025), Performance Audit on Functioning of Delhi Jal Board (Report No. 3 of 2025), p. 14.
23. CAG Report on Functioning of Delhi Jal Board Submitted in Assembly, March 2026
24. Gender Equality And Empowerment Of Women For Urban Planning
25. Rajya Sabha, Unstarred Question No. 3741, Ministry Of Environment, Forest And Climate Change, Government of India, 2025

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Uncertainty about delivered water quality then deters the very off takers whose demand would make that network bankable. Therefore, adoption remains highly uneven across cities and states.

Haryana uses only about 18 per cent of its treated wastewater capacity,[26] while Chennai recycles around 90 MLD of treated wastewater, equivalent to roughly ten per cent of the city's total water supply, through dedicated tertiary treatment plants serving industrial users.[27] AMRUT 2.0's emphasis on City Water Balance Plans, treated-water reuse, and circular water management is intended to close this gap, but the contrast between cities shows that India's transition from wastewater disposal to wastewater reuse is still at an early stage.

Governance and financial sustainability

Ultimately, many of these challenges converge at the governance bodies/ institutions – like ULBs, districts, GPs, villages, which are increasingly central to delivering India's water-security ambitions. In some states, however, responsibilities remain shared across multiple institutions. In Uttar Pradesh, for example, major planning and implementation functions have been carried out by the Jal Nigam, while ULBs accounted for around 43-45 per cent[28] of related expenditure, reflecting a model where state-level agencies continue to play a significant delivery role alongside city governments.

Financially, ULBs are being asked to manage an expanding asset base with relatively limited own-source revenues. Municipal corporations' own revenue receipts amount to about 0.6 per cent of GDP, compared with ~9 per cent for the Centre and ~15 per cent for the states. Revenue capacity is also concentrated, with the 10 largest municipal corporations accounting for more than 58 per cent of municipal revenue nationally, highlighting the differing capabilities of large metros and smaller cities. [29]

New financing mechanisms are beginning to emerge but remain at an early stage. Over a decade of AMRUT, 13 ULBs raised INR 4,984 crores through municipal bonds, while the programme itself sanctioned projects worth INR 2.73 lakh crore, indicating that public investment continues to be the dominant funding source for urban water infrastructure. [30]

While rural water financing remains

predominantly public, the existence of alternative channels is still at an early stage. JJM was launched with an initial outlay of INR 3.60 lakh crore, including INR 2.08 lakh crore Central share, almost fully utilised by 2024-25.[31] Complementary mechanisms such as the Finance Commission tied grants, CSR, community contribution, DMF convergence and Rashtriya Jal Jeevan Kosh emerged as additional financing sources but remain supplementary to Centre-State budgetary financing.

At the same time, the scale of assets being created is substantial. AMRUT 2.0 approved INR 91,557 crore of water-supply projects and nearly 89,594 km of distribution network[32], significantly expanding the infrastructure available to cities and citizens. As this network grows, ensuring adequate operational capacity, sustainable financing, and long-term asset management is expected to become increasingly important. The opportunity is considerable: stronger local institutions can translate these investments into more reliable water quality, improved service continuity, and broader access, including in rapidly growing peri-urban areas and underserved communities through aligning investments with viable O&M and revenue models, and readjusting tariffs, reuse contracts or PPP/SPV structures where needed. This can help control long-term liabilities, improve investor confidence, and translate investments into better water quality, reduced losses, improved service continuity, and wider access.

Water services possess a unique and often overlooked characteristic of a natural monopoly: infrastructure may exist, yet service delivery can remain absent. Pipelines, treatment plants, and networks can be built, but reliable, safe, and sustainable water service is not guaranteed. This paradox lies at the heart of the sector's challenges and should be the central focus of investment strategies, which must prioritise service outcomes and institutional performance rather than infrastructure creation alone. In other words, India's next water challenge is less about building infrastructure and more about ensuring that local institutions have the capabilities, resources, and governance arrangements needed to operate, maintain, and optimise the assets being created.

26. IWRAP 2025-2027; Efficiency of non-revenue water reduction in improving water supply performance in Indian metropolises, IWA, March 2023
27. Water in Circular Economy and Resilience (WICER), The Case of Chennai, World Bank Group, India
28. Performance Audit Report on Implementation of 74th Constitutional Amendment Act, Comptroller and Auditor General of India
29. Fiscal Position of Municipal Corporations, November 2024
30. 10 Years of AMRUT, PIB Backgrounder, June 2025
31. Assessment of Jal Jeevan Mission, Ministry of Jal Shakti, Press Information Bureau, February 2026
32. PROJECTS UNDERWAY UNDER AMRUT 2.0, Press Information Bureau, July 2023

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2.4 A value chain response for closing the gap

Closing the convergence gap requires action across multiple stages of the water value chain, from the sources that supply water, to the infrastructure that moves and treats it, to the services delivered to citizens, and finally to the institutions that govern and finance the system.

A. Upstream - protecting the source

Shift from infrastructure planning to water allocation planning

Investment continues to focus on reservoirs, pipelines and treatment assets, often without addressing how scarce water could be allocated among competing users. What is increasingly needed is a basin-level decision system that integrates water availability, groundwater, climate risks, water quality and sectoral demand to guide allocation, manage trade-offs and anticipate future stresses.

Treat aquifers as managed assets rather than natural endowments

Groundwater management has largely focused on recharge, even though recharge alone cannot ensure sustainability if extraction remains unchecked. A more effective approach is to manage aquifers as strategic assets by tracking recharge, withdrawals, storage and safe yield limits. Gram Panchayat-level water budgeting can align groundwater availability with future demand, while digital tools can support early action by identifying emerging deficits and guiding interventions before water stress becomes a crisis.

Institutionalise source protection as a permanent service function

Source sustainability must become a continuous operational responsibility rather than a one-time project activity. Many institutions are designed to manage water supply infrastructure, not the sources that sustain it. Dedicated source stewardship functions within local water institutions can strengthen accountability for monitoring source health, maintaining recharge assets and identifying emerging risks. Simple source health scorecards tracking quantity, quality and seasonal reliability can provide early warning signals and enable timely interventions before service disruptions occur.

Control pollution through catchment accountability rather than treatment assets alone

Pollution control should focus on river and watershed outcomes, not treatment infrastructure alone. River health depends on managing pollutant loads across the entire catchment and maintaining accountability for all pollution sources. Treatment plants, CETPs and zero-liquid-discharge systems need to operate within basin-wide pollution management frameworks linked to measurable water-quality outcomes. Effective governance requires real-time monitoring, enforceable discharge standards and clear accountability for dischargers. As digital monitoring systems become more widespread, compliance can shift from periodic inspections to continuous oversight, enabling early intervention before pollution reaches rivers and water bodies.[33]

B. Midstream - infrastructure and asset creation

Move from asset creation to service capacity

The sector continues to reward infrastructure creation rather than actual service delivery. Many cities possess treatment plants, reservoirs, and extensive networks that operate well below their design potential. The critical constraint is often not a lack of assets but poor utilisation, weak operations, and fragmented system management. Infrastructure investments should therefore be assessed against their ability to improve service outcomes such as hours of supply, water quality, treatment compliance, and network reliability. The real objective is to increase service capacity from the system as a whole, not simply to add more physical capacity through new projects.

Design infrastructure as resource factories

Water and wastewater facilities are still planned primarily as compliance assets that deliver safe disposal. This overlooks their potential as producers of valuable resources. Sewage contains water for industrial reuse, nutrients, biosolids, biogas, and carbon-reduction benefits that can create long-term revenue streams. Infrastructure planning should start by identifying potential off-takers and markets for these resources before determining plant size and technology. In the future, the financial sustainability of utilities may depend less on tariffs and more on their ability to monetise recovered water, energy, nutrients, and environmental attributes.

33. IN THE SUPREME COURT OF INDIA INHERENT/CIVIL APPELLATE JURISDICTION SUO MOTO WRIT PETITION (CIVIL) NO(S). 8 OF 2025, Supreme Court of India, 2025

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Decouple service coverage from sewer coverage

Urban sanitation planning often assumes that complete sewer network coverage is the only pathway to safe sanitation. In practice, network expansion frequently lags urban growth by decades, leaving large populations dependent on informal systems. The more important objective is universal treatment rather than universal sewer connectivity. Faecal sludge treatment, co-treatment at existing STPs, decentralised systems, and cluster-level sanitation infrastructure can deliver environmental outcomes comparable to conventional sewerage at significantly lower cost and shorter timelines. Cities should prioritise the fastest route to safe treatment rather than committing exclusively to fully centralised systems.

Build for failure rather than normal operations

Many utilities are designed around average operating conditions even though service disruption increasingly results from droughts, floods, power outages, and infrastructure failures. Resilience should therefore be treated as a core design requirement rather than a contingency measure. Redundant intakes, backup transmission routes, emergency power, mobile treatment systems, and mutual-aid arrangements can ensure continuity during shocks. The true test of infrastructure is not how efficiently it performs under normal conditions, but how effectively it continues delivering critical services when conditions become abnormal. Infrastructure should be designed with disruption as an expectation, not an exception.

Treat energy as a strategic design variable

Energy is commonly viewed as an operational expense to be managed after commissioning. However, treatment technologies, pumping requirements, hydraulic design, and process selection lock utilities into decades of energy consumption. Energy efficiency should therefore be embedded into infrastructure planning as a primary decision criterion.

Utilities that reduce energy demand first are better positioned to adopt renewable energy, lower operating costs, improve financial sustainability, and reduce exposure to power-price volatility. The cheapest unit of energy remains the one that never needs to be consumed.

Invest in operational intelligence

One of the sector's least recognised gaps is that utilities often invest heavily in physical infrastructure while underinvesting in the systems needed to operate it effectively. Tools such as SCADA, district metered areas, pressure monitoring, smart metering, GIS-linked asset management, and predictive maintenance frequently generate higher returns than additional civil works. These systems increase asset utilisation, reduce losses, improve reliability, and allow problems to be addressed before they become failures. Future infrastructure programmes should view operational intelligence as essential infrastructure because the performance of assets increasingly depends on the quality of information managing them.

C. Downstream - service delivery, reuse and circularity

Shift from infrastructure access to service outcomes

The water sector has largely succeeded in extending physical connectivity, but access to infrastructure does not automatically translate into reliable service. Millions of households are connected to networks that still deliver intermittent supply, low pressure, poor quality, or uncertain reliability. The next challenge is therefore to move beyond connection-centric targets and focus on measurable service outcomes such as continuity, pressure, quality, responsiveness, and equity. Utilities should be evaluated based on the customer experience they deliver, not the assets they own. The true measure of success is whether water arrives when needed, in adequate quantity and quality, regardless of location or income.

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Treat consumers as participants, not beneficiaries

Many water systems are designed around engineering assets rather than customer relationships. Utilities often know more about their pumps and pipelines than about consumer expectations, service failures, or demand patterns. Customer service, grievance management, communication, and transparency should therefore be treated as core utility functions rather than administrative add-ons. Digital platforms, service-level agreements, real-time dashboards, and structured feedback loops can convert customer interactions into operational intelligence. Households should not merely receive services; they should become active participants whose usage patterns, complaints, and preferences directly influence planning, maintenance, and service improvement decisions.

Create markets for treated wastewater, not just treatment plants

Many wastewater investments continue to be justified by environmental compliance, which makes treatment an ongoing cost centre. A more sustainable approach is to treat wastewater as a resource with economic value. The constraint in many cities is not treatment capacity but the absence of reliable demand for treated water. Reuse planning should therefore begin with identifying industrial, agricultural, municipal, or commercial off-takers before infrastructure is designed. Long-term purchase agreements, reuse mandates, and differentiated pricing can transform wastewater from a disposal problem into a revenue-generating asset, improving both utility finances and water security.

Use metering to manage demand, not just billing

Water metering is often introduced as a billing tool, yet one of its greatest value lies in generating information. Without accurate consumption data, utilities are effectively managing blind systems where losses, demand patterns, leakage, and inefficiencies remain hidden. Smart metering can provide a foundation for demand management, infrastructure planning, source sustainability, and service optimization. The objective is not merely to charge customers more accurately, but to understand how water moves through the system and how consumers use it. Data generated from metering should become a strategic planning resource rather than a narrow revenue-management instrument.

D. Cross-cutting enablers - utility reform, finance, capacity and digital operations

Reform utilities around accountability, not administration

Many water institutions remain organised as government departments where accountability is fragmented across planning, construction, operations, and regulation. As a result, no single entity is responsible for outcomes such as service reliability, water quality, financial sustainability, or source protection. The sector's long-term challenge is therefore institutional as well as technical. Utilities should be transformed into professionally managed, ring-fenced service organisations with clear mandates, measurable performance indicators, and end-to-end responsibility across the source-to-tap-to-reuse value chain. The focus should shift from managing projects and budgets to delivering accountable, outcome-oriented public services.

Recognise that finance follows credible institutions

The water sector's financing challenge is often portrayed as a shortage of money, when the more fundamental constraint is the lack of investable institutions. Capital is generally available for projects, but investors and lenders struggle to identify utilities with predictable revenues, reliable governance, and clear accountability. Sustainable financing therefore requires moving beyond grant-dependent, asset-creation models toward performance-linked funding structures. Tariff reforms, blended finance, hybrid-annuity mechanisms, and outcome-based payments can succeed only when supported by institutions capable of managing commercial risks, honouring contracts, and demonstrating measurable service improvements over time.

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Build water as a profession, not just a sector

Unlike many other professional/services fields, the water sector lacks a clearly defined professional pathway that young professionals can aspire to at the beginning of their careers. Talent enters through engineering, environmental science, urban planning, public policy, operations, finance, or research, but there is rarely a structured progression connecting these disciplines into a recognised profession. Consequently, utilities often struggle to attract, develop, and retain high-quality talent. The sector must invest in career frameworks, role-based certification, specialised training, utility leadership programmes, and centres of excellence that create a visible and respected professional identity for water practitioners throughout their careers.

Make water systems observable

One of the least appreciated weaknesses of many utilities is that they operate largely without real-time visibility of what is happening in their own networks. Decisions are often based on periodic reporting, complaints, or manual inspections rather than continuous operational intelligence. As systems become larger and more complex, this approach becomes increasingly ineffective. Utilities require the ability to monitor flows, pressures, water quality, assets, energy use, and service performance in real time. The future of water management depends on transforming utilities from reactive organisations that respond to failures into observable systems that detect risks, inefficiencies, and service disruptions before they occur.

Use artificial intelligence to augment operational judgment

One of the greatest values of AI in water management is unlikely to come from automation alone, but from helping utilities make better operational decisions. Water systems generate large amounts of data that often remain underutilised because utilities lack the capacity to interpret them at scale. AI and machine-learning tools can identify hidden leakage patterns, predict equipment failures, forecast demand, optimise pumping schedules, and detect water-quality anomalies faster than conventional approaches. The objective is not to replace operators but to augment human judgment, enabling smaller teams to manage increasingly complex systems with greater speed, accuracy, and consistency.

Shift from reactive maintenance to asset stewardship

Utilities traditionally invest heavily in building assets and relatively little in understanding their condition over time. As infrastructure ages, this creates growing maintenance backlogs, unexpected failures, and escalating replacement costs. A more sustainable approach is to treat assets as long-term portfolios that require continuous stewardship throughout their lifecycle. GIS-linked asset registers, condition assessments, predictive maintenance tools, and risk-based renewal planning can help utilities prioritize interventions before failures occur. The future challenge is not constructing more infrastructure but preserving and extracting maximum value from the substantial infrastructure that already exists.

Water security, on its own, is not an end point. Once water is secured at source, treated efficiently in the middle, and returned to productive use at the end, the same logic that governs the water chain protect the input, minimise the loss, recover the residue - becomes the operating principle for India's wider environmental infrastructure. Treated used water becomes an industrial feedstock; sewage sludge becomes energy and nutrient recovery; the discipline of measuring what enters and leaves a system carries directly into solid waste. This is why water cannot be planned in isolation from waste, or from the circular economy, as they share the same infrastructure, the same municipal balance sheet, the same institutional capacity, and ultimately the same test - whether India's growth is built on resources it consumes once, or on resources it keeps in circulation.

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Water and wastewater technology choices are often evaluated on treatment efficiency, energy consumption, nutrient recovery, or lifecycle costs. However, technologies do not operate in isolation; they must function within the institutional, financial, and operational realities of the implementing agency. A solution that offers superior long-term economics may still fail if it requires highly specialised operators, intensive process control, complex maintenance regimes, or risk management capabilities that utilities do not possess. Technology selection should therefore be based not only on technical performance, but also on investment readiness, operational sustainability, land availability, financing structures, regulatory requirements, and local capacity. This requires utilities, regulators, financiers, and approving authorities to adopt multidisciplinary decision-making frameworks that combine engineering, finance, operations, environmental management, risk assessment, and organisational capability. The most appropriate technology is not necessarily the most advanced one, but the one that can be sustainably operated, maintained, and scaled over its entire lifecycle. The applicable manuals, processes, benchmarks, and approval processes must evolve rapidly to accommodate this requirement of infrastructure creation and operation.



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Section 3:

Solid Waste Management, from policy to the pavement



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India has come a long way in Solid Waste Management since the inception of Swachh Bharat Mission (urban) which is considered the world's largest Waste Management Movement. India crossed a regulatory threshold on 1 April 2026. The Solid Waste Management Rules, 2026, notified by the Ministry of Environment, Forest and Climate Change on 28 January 2026, replaced the decade-old 2016 framework and folded circular economy and Extended Producer Responsibility principles directly into municipal waste governance[34]. On the same day that the new Rules came into force, the Supreme Court was already supervising their implementation through a pan-India order in Civil Appeal No. 6174 of 2023, turning what used to be a municipal housekeeping function into a matter of constitutional oversight[35].

This section examines four critical challenges shaping India's solid waste management sector: increasing compliance obligations and regulatory oversight, persistent gaps in last-mile service delivery, underutilised opportunities for resource recovery and circularity, and execution constraints that cannot be resolved through policy reforms alone.

3.1 SWM rules 2026: The compliance clock of ULB and industry player is now racing

The SWM Rules, 2026 supersede the SWM Rules, 2016 under the Environment (Protection) Act, 1986 and set a steep hard compliance timeline. CPCB's 2023-24 baseline puts India's daily municipal solid waste generation at approximately 1.85 lakh tonnes, of which residential societies, government buildings and universities together account for nearly 30 per cent, making bulk institutional generators a central, not peripheral, compliance audience[36].

What's changed: key amendments and new obligations for ULBs and bulk generators

The 2026 Rules move India from two-stream to four-stream segregation at source, and widen the compliance net for bulk generators through a new Extended Bulk Waste Generator Responsibility (EBWGR):

- Wet waste (kitchen waste, food leftovers, fruit and vegetable peels) must be composted or bio-methanated at the nearest facility
- Dry waste (plastic, paper, metal, glass, wood, rubber) must be routed to Material Recovery Facilities for sorting and recycling
- Sanitary waste (diapers, sanitary products) must be securely wrapped and stored separately, and special care waste (bulbs, batteries, medicines) handed over only to authorised agencies
- EBWGR requires bulk generators to process wet waste on-site, or obtain a responsibility certificate where on-site processing is not feasible
- A Refuse Derived Fuel mandate requires cement and waste-to-energy plants to raise RDF usage from 5 per cent to 15 per cent over six years.

A centralised online portal, developed and administered by CPCB, now could track waste generation, collection, transportation, processing and disposal end to end, replacing the physical, paper-based reporting that made verification nearly impossible under the 2016 regime[37]. The Rules also formalise a waste hierarchy of prevention, reduction, reuse, recycling and recovery, with disposal to landfill positioned explicitly as the last resort [38].

34. Press Information Bureau, Government of India, "New Solid Waste Management Rules Notified; To Come into Force from April 1, 2026," Ministry of Environment, Forest and Climate Change, 28 January 2026.
35. Verdictum, "'Waste-Free India'; Non-Compliance With Solid Waste Management Rules, 2026 Not A Mere Administrative Lapse: Supreme Court Mandates Three-Tier Enforcement," Bhopal Municipal Corporation v. Dr Subhash C. Pandey, Civil Appeal No. 6174/2023, 23 February 2026.
36. Drishti IAS, "Solid Waste Management (SWM) Rules, 2026," citing CPCB data for 2023-24 (India generates approximately 1.85 lakh tonnes per day of solid waste).
37. Down To Earth, "India's 2026 Waste Management Rules: Digital Traceability and Compliance Challenges," 16 March 2026.
38. Sanskriti IAS, "Solid Waste Management Rules, 2026: Towards a Circular and Decentralised Urban Waste System."

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Enforcement mechanisms: penalties, timelines, and monitoring under the new framework

Non-compliance now mandates environmental compensation under the 'polluter pays' principle, covering operation without registration, false reporting, submission of forged documents, and improper solid waste management practices generally; CPCB frames the guidelines while State Pollution Control Boards and Pollution Control Committees levy the compensation. Landfills face parallel tightening: only non-recyclable and inert waste may be sent to landfill, unsegregated waste attracts higher fees, and every landfill is now subject to mandatory annual audits with District Collectors monitoring operations directly[39].

On legacy waste, every existing dumpsite must be geographically mapped and evaluated by October 2026, with time-bound biomining and bioremediation mandated thereafter, and a graded, faster land-allocation process introduced for new processing and disposal facilities, paired with mandatory buffer zones for larger sites[40].

Implications of Supreme Court's directions

The compliance conversation for ULBs and bulk generators cannot be read from the Rules alone. In a related recent case, the Supreme Court held that non-compliance with the SWM Rules, 2026 is not a mere administrative lapse, linking municipal waste governance directly to the Article 21 right to a clean and healthy environment, and mandating a three-tier enforcement structure running from the Union ministries down to Block and Ward-level nodal officers. On 25 May 2026, the Court went a step further and constituted a Supreme Court-empowered Monitoring Committee, comprising five Union Secretaries and the CPCB Member Secretary, with authority to supervise, monitor and issue directives for implementation of the 2026 Rules nationwid[41].

For programme teams, two directional details from the Court's 5 May 2026 order matter more than the headline compliance date. First, the Court has explicitly endorsed a Special Purpose Vehicle model for SWM, under which ULBs retain primary collection and transportation while a specialised SPV runs processing across all waste streams against well-defined KPIs, with collection and transportation also transferable to the SPV where appropriate; this has a direct bearing on how Technical Support Units and institutional design should be approached[42]. Second, as of the Court's order, only around 1,700 ULBs nationally had registered on CPCB's EPR portal, a number the Court itself flagged as inadequate against the scale of the compliance base and a useful proxy for how far implementation still has to travel behind the policy text.

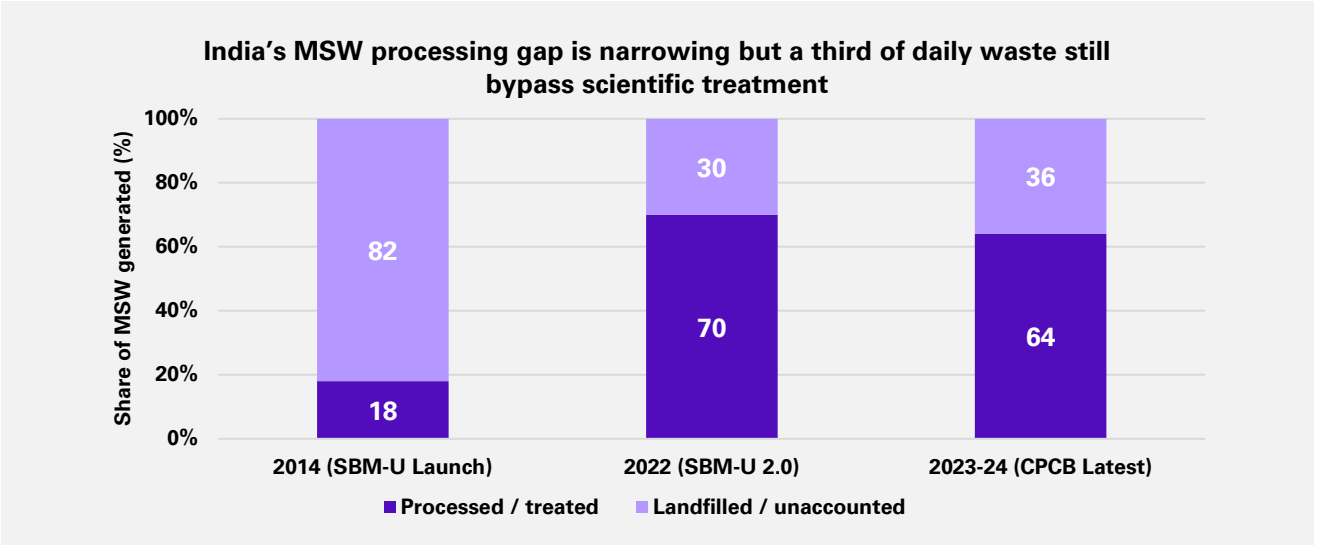


Figure 3.1: India's MSW processing gap, CPCB estimates, FY 2013-14 to FY 2023-24 {{28,38}}

39. PSR Compliance, "Solid Waste - SWM Rules 2026 Update," 9 April 2026.
40. Lawrbit, "Solid Waste Management Rules 2026: Mandatory Changes and Rules," 13 February 2026; Insights on India, "Solid Waste Management (SWM) Rules, 2026," 31 March 2026.
41. K.B.S. Sidhu, "From Rules to Reckoning: The Supreme Court Takes Charge of India's Garbage Crisis," 24 June 2026, on the Supreme Court-empowered Monitoring Committee constituted 25 May 2026 in Civil Appeal No. 6174/2023.
42. Supreme Court of India, Order dated 5 May 2026, Civil Appeal No. 6174/2023 (on Special Purpose Vehicle design for SWM and CPCB EPR portal registration by Urban Local Bodies).

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3.2 The last mile problem: why segregation and collection still break at scale

SBM 2.0's Urban dashboard reports door-to-door collection at 97 per cent nationally and source segregation at 90 per cent, yet independent assessments tell a different story: a World Bank study found around 77 per cent of mixed waste still ends up in landfills, and significant proportion of waste unaccounted for[43]. CPCB's more recent estimate puts processed waste at 64 per cent of the daily stream, with the remaining 36 per cent, roughly 4,241 tonnes a day, going to unsanitary landfills or informal dumpsites[44]. The Solid Waste Management Bill, 2025, introduced in the Rajya Sabha on 5 December 2025, states plainly that source segregation remains below 50 per cent in major cities despite a decade of EPR and digital compliance mechanisms[45].

Source segregation: why compliance stalls below 60 per cent outside metros

The gap is not uniform across the urban hierarchy, and averaging it away is itself part of the problem. Cities with populations above 50 lakh report an average collection rate of 68.5 per cent, against 96.7 per cent for cities under 10 lakh; segregation follows the same pattern, practised in 38 per cent of large cities against 54 per cent of small cities[46]. Even these self-reported figures should be read with caution: a CAG performance audit found that none of the test-checked ULBs had followed a systematic procedure for estimating average waste generation, challenging the basis on which processing capacity is planned in the first place.

The picture is not uniformly weak though: for example, Pune has sustained reported source segregation above 95 per cent, and Visakhapatnam had improved segregation by 38 per cent points between 2020 and 2023, evidence that the constraint is institutional and behavioural rather than just technical[47].

Collection and transportation: fleet, route, and IEC gaps

Larger cities carry a structurally harder collection problem: more informal settlements, denser traffic-constrained routes, and bulk generators spread across a wider geography. All of these strain fleet planning and route optimisation that were designed around 2016-era volumes. The absence of reliable, sample-based waste generation data compounds this, since fleet sizing, route frequency and MRF siting decisions are all being made on inputs that CAG has already flagged as unsystematic in test-checked ULBs.

The four-stream mandate under the 2026 Rules effectively doubles the collection complexity that ULB fleets and transfer stations were built for, from two-stream to four-stream handling, without a commensurate national programme yet visible for fleet retrofitting or transfer station redesign.

Behaviour change at scale: what's worked, and what hasn't

Cities that have moved segregation compliance durably above 90 per cent, Indore and Surat among them, share a common pattern: a dedicated IEC and BCC cell inside the ULB, transparent public reporting of generation, segregation, processing and penalty data, and consistent citizen-facing communication through multiple channels rather than a single campaign moment.

Where this institutional muscle is absent, the star-rating and Garbage Free City certification incentives built into Swachh Survekshan have proven necessary: the 23-indicator assessment framework has driven visible investment in infrastructure.

43. Down To Earth, "Fostering behaviour change in solid waste management: Here are some essential prerequisites for cities," citing SBM 2.0 Urban dashboard, a World Bank study, and CPCB (2020-21) data.
44. Scroll.in, "Four-way segregation, landfill restrictions: Will India's new solid waste rules bring about change?," 18 March 2026, citing Central Pollution Control Board data.
45. IndiaLaw, "The Solid Waste Management Bill, 2025: Establishing A Statutory Framework For Segregation, Recycling And Sustainable Municipal Solid Waste Governance In India," 27 January 2026.
46. PubAdmin Institute, "Effective Strategies for Municipal Solid Waste Management in India," 18 April 2026.
47. Ecoinsee Journal, "Scenario of Solid Waste Management in Indian Cities" (case studies of Pune, Tirupati and Visakhapatnam).

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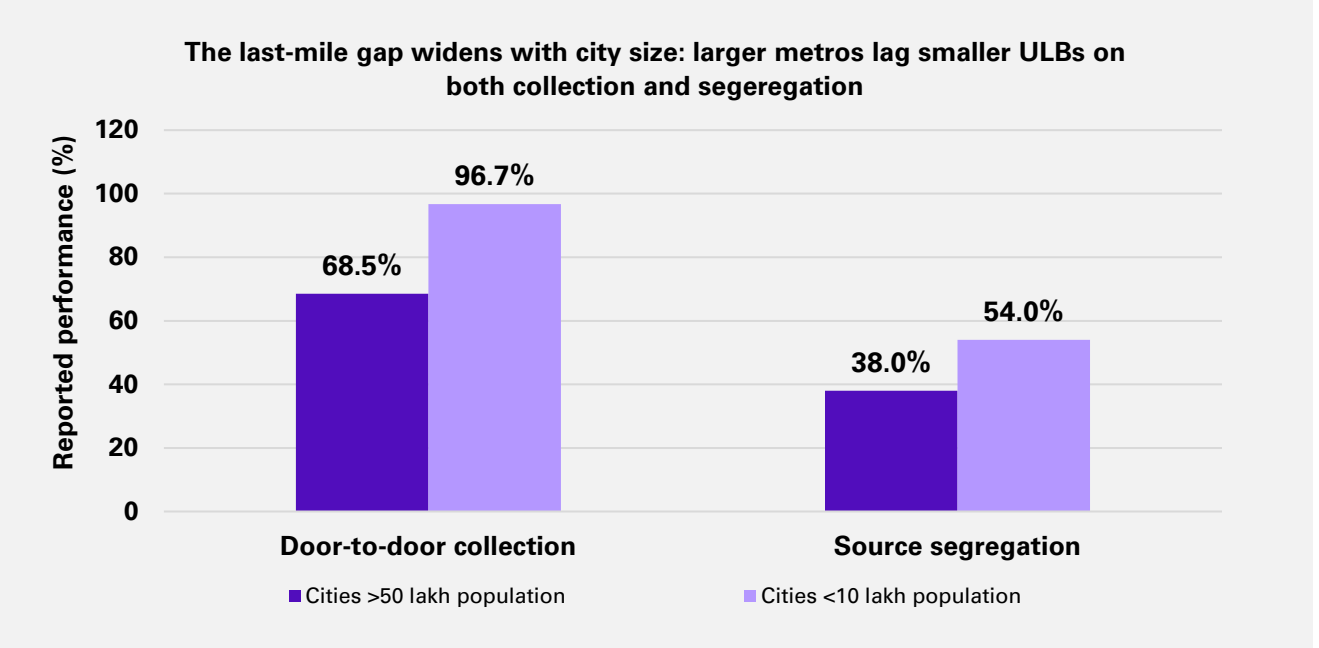


Figure 3.2: Reported collection and segregation performance by city size {{36}}

3.3 From dump to asset: waste-to-energy, biomethanation, and land-based recovery

The 2026 Rules position landfills as a last-resort disposal option, not the default solution. The infrastructure needed to support this shift, including waste-to-energy plants, biomethanation facilities, Material Recovery Facilities (MRFs) and dumpsite remediation, is expanding across India. However, progress remains uneven across technologies, cities and states.

Biomethanation and bio-CNG: emerging models

India added 228 MWe of dedicated waste-to-energy capacity over the past decade, alongside 2,361 MW of biomass capacity, under the National Bioenergy Programme, which carried a budget of INR 998 crore for the 2022-23 to 2025-26 period[48].

The RDF mandate under the 2026 Rules, requiring cement and WtE plants to enhance RDF utilisation from 5 per cent to 15 per cent over six years, is designed precisely to close the feedstock-quality gap that has historically constrained WtE viability beyond a handful of well-run plants.

The GOBARdhan initiative reported 189 functional Compressed Biogas projects and 979 functional community or cluster biogas plants nationally as of 27 January 2026, with Uttar Pradesh leading on CBG installations and Chhattisgarh leading on community biogas plants, and adoption visible across Madhya Pradesh, Gujarat, Maharashtra, Karnataka, Haryana and Tamil Nadu rather than concentrated in one region[49].

Advanced waste processing: MRFs, RDF, and mechanical-biological treatment

Dry waste under the 2026 Rules must flow through Material Recovery Facilities for sorting and onward recycling, positioning MRF capacity as the connective tissue between source segregation and the EPR value chains.

The RDF ramp-up mandate effectively guarantees offtake for cement co-processing over the next six years, which should, in principle, de-risk investment in MRF and mechanical-biological treatment capacity, provided ULBs can deliver the segregated, contamination-controlled dry waste stream that RDF-grade feedstock requires.

48. India Brand Equity Foundation (IBEF), "India Adds 2,361 MW Biomass Capacity, 228 MWe Waste-to-Energy and 2.88 Lakh Biogas Plants in Last Ten Years," 18 December 2025.

49. Organiser, "GOBARdhan: Turning waste into wealth and transforming the rural economy of Swachh Bharat," 5 March 2026, citing state-level GOBARdhan portal data as of 27 January 2026.

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Legacy dumpsite remediation and land recovery as an asset class

India has identified 2,479 legacy dumpsites nationally, holding an estimated 25 crore metric tonnes of waste across roughly 15,000 acres; the Dumpsite Remediation Accelerator Programme, launched in November 2025, has prioritised 214 high-impact sites accounting for nearly 80 per cent of the remaining legacy volume, about 8.6 crore metric tonnes, with a Lakshya Zero Dumpsites target of October 2026[50]. National progress in 2025 alone saw 459 dumpsites across 438 cities in 26 states fully remediated, adding 183 lakh metric tonnes to a cumulative total of 1,138 dumpsites remediated across 1,048 cities in 29 states, representing 877 lakh metric tonnes since the programme began[51]. SBM-U 2.0's overall target is remediation of 16 crore tonnes of legacy waste and reclamation of over 15,000 acres, with Delhi's Bhalswa, Ghazipur and Okhla sites alone drawing allocations exceeding INR 776 crore[52].

Reclaimed land is earmarked, in order of preference, for SWM infrastructure or green cover, effectively converting a decades-old liability into a pipeline of urban land assets.

Figure 3.3: Legacy waste biomined at Delhi's three major landfill sites {{17}}

Recent policy support has strengthened the business case for resource recovery. Improved CBG pricing and GOBARdhan incentives are enhancing project viability and enabling revenues to support a larger share of collection, processing and recovery infrastructure. However, long-term success depends on more than technology or financing. Viability requires strong local ecosystems that connect feedstock supply, processing, offtake markets and end-use demand. As market conditions evolve, cities need access to technical expertise and transaction support to rapidly design, test and adapt business models. The strategic lesson is clear: infrastructure creates assets, but systems create value. The greatest opportunities emerge when waste, water, energy and land are managed as interconnected resource flows rather than separate sectors.

3.4 The execution gap: why implementation, not policy, is India's real SWM bottleneck

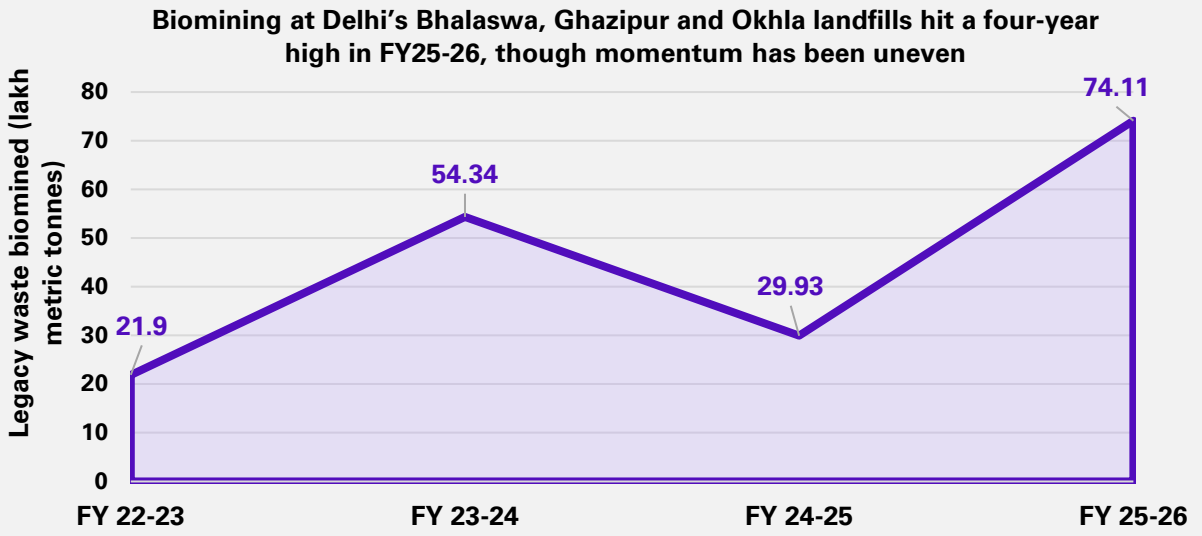
Every layer of India's SWM framework, the 2026 Rules, the Supreme Court's monitoring architecture, and the financing instruments described below, is now converging on the same conclusion: the binding constraint has shifted from policy design to institutional delivery capacity.

Institutional capacity: staffing, technical skills, and ULB readiness

Financial and technical capacity constraints in ULBs, and weak accountability mechanisms for bulk waste generators, are cited explicitly in the Statement of Objects and Reasons for the Solid Waste Management Bill, 2025 as a primary reason the existing framework has fallen short despite progressive provisions on EPR and digital compliance. That capacity gap is precisely why, as of the Supreme Court's own review, only around 1,700 ULBs nationally had completed registration on CPCB's EPR portal, a leading indicator that tracks institutional readiness more directly than headline collection or segregation percentages.

50. GKToday, "India Targets Zero Dumpsites Under DRAP by 2026," 2 February 2026.
51. Press Information Bureau, Government of India, "Lakshya Zero Dumpsite: India's Drive to Eliminate City Dumpsites Under Swachh Bharat," 31 January 2026.
52. Current World Environment journal, "Sustainable Management of Landfill Sites in India: Addressing Environmental, Health, and Socioeconomic Challenges," 2025, on SBM-U 2.0 legacy waste remediation targets.

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Structuring institutions for delivery, not just compliance

The Supreme Court's 5 May 2026 order in Civil Appeal No. 6174 of 2023 does not merely direct compliance; it prescribes an institutional model. ULBs are directed to retain primary collection and transportation as their core mandate, while a dedicated Special Purpose Vehicle, specialised in processing across all waste streams, takes on plant operations against well-structured KPIs, with the option to transfer collection and transportation to the SPV as well where that improves delivery.

This is a meaningful shift: it moves the unit of accountability from a generalist municipal engineering department to a specialised, KPI-governed entity, and it gives state governments and Technical Support Unit teams judicial cover to pursue SPV or hybrid-annuity structures that might otherwise have faced governance challenges at the ULB level.

PPP integration: risk allocation and private participation in SWM infrastructure

Private participation in India's urban infrastructure has historically remained relatively limited, with only around 6 per cent of urban infrastructure projects adopting PPP structures and financial closure achieved in about 12 per cent of cases by 2023. This highlights the scope for further strengthening project financing frameworks, streamlining viability gap funding processes, and deepening municipal capital market participation to unlock greater private-sector involvement.[53]

SWM-specific PPP models, from processing-plant concessions to SPV-based operate-and-maintain contracts, fall squarely inside the fix the Urban Challenge Fund is designed to catalyse, by tying central assistance to demonstrated market participation rather than releasing it as an unconditional grant.

Financing execution: DPRs, viability gap funding, blended capital, and the Urban Challenge Fund

The Urban Challenge Fund, announced in the Union Budget 2025-26 and approved by Cabinet with a central outlay of INR 1,00,000 crore for FY 2025-26 to FY 2030-31 (extendable to FY 2033-34), is explicitly structured to catalyse roughly INR 4 lakh crore of investment over five years by conditioning central assistance on market participation.[54] The Centre funds up to 25 per cent of a bankable project's cost, provided at least 50 per cent is mobilised from market sources, municipal bonds, bank loans or PPPs, with the balance contributed by states, UTs or ULBs; water and sanitation, explicitly including integrated solid waste management and legacy waste remediation, is one of three thematic pillars the Fund covers.

53. GKToday, "Urban Challenge Fund," 21 August 2025, citing pre-UCF PPP uptake and financial closure baselines.

54. Drishti IAS, "Urban Challenge Fund," 16 February 2026; Press Information Bureau, "Cabinet approves Rs. One Lakh Crore Urban Challenge Fund to Drive Market-Led Urban Transformation."

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Programme	Outlay/scale	SWM-relevant design feature
SWM Rules, 2026	Nationwide, effective 1 Apr 2026	Four-stream segregation, EBWGR, RDF ramp 5% to 15% over six years
Dumpsite Remediation Accelerator Programme	214 priority sites of 2,479 identified	Targets 8.6 crore MT (80% of remaining legacy waste) by Oct 2026
Urban Challenge Fund	INR1,00,000 crore, FY26 to FY31 (extendable to FY34)	25% central assistance, min. 50% market mobilisation, outcome-linked
Credit Repayment Guarantee Scheme	INR5,000 crore corpus	Guarantees up to 70% of first loan for small / hill-state ULBs
SC Monitoring Committee (CA 6174/2023)	5 Union Secretaries + CPCB Member Secretary	Pan-India supervision; SPV model for SWM processing endorsed

Table 3.1: The stacked policy, judicial and financing architecture now governing SWM execution in India (49)

3.5 Beyond compliance: The five priorities for India's next SWM decade

India's solid waste management challenge is no longer a policy-design problem. The SWM Rules, 2026 are comprehensive, the Supreme Court's monitoring architecture is active, and the financing instruments, from the Urban Challenge Fund to the Credit Repayment Guarantee Scheme, now exist on paper. What determines whether India's cities look meaningfully different by 2030 is a sharper and harder question: whether six groups of stakeholders, ULBs, state governments, Technical Support Units, the private sector, investors and technology providers, change how they operate, not just what they comply with.

ULBs: from operators to contract managers

For most of the last decade, ULBs have tried to do everything themselves: collect, transport, process and monitor, often with the same generalist municipal engineering department responsible for all four. The Supreme Court's own SPV directive points to a different model, one where the ULB owns service outcomes, a specialised private or SPV operator owns operational efficiency, and the ULB owns monitoring against KPIs. The practical implication for the next decade is a reallocation of municipal capacity: less time spent running plants and fleets directly, more time spent writing, awarding, and holding counterparties to outcome-based contracts. ULBs that make this shift first can free up scarce technical staff for exactly the governance functions, segregation enforcement, tipping-fee collection, contract management, that this section has identified as the actual binding constraint.

State governments: from city-by-city plans to regional waste economies

Not every town of 80,000 people needs its own waste-to-energy plant, biomethanation facility, RDF line and recycling park. Yet, SWM planning largely proceeds city by city owing to the governance structure, replicating the same infrastructure component requirements. The more defensible model, and one several states are already exploring informally, is the regional waste cluster: a group of neighbouring ULBs sharing a common processing facility, MRF or biomining contractor, sized to the combined feedstock of the cluster rather than any single town's demand. This is a straightforward extension of how AMRUT and industrial corridor planning already work at state level, and it directly addresses the sub-scale investment problem that has left smaller ULBs either under-served or saddled with processing capacity they cannot fill. States that formalise cluster boundaries and shared-facility financing now can be the ones positioned to draw down Urban Challenge Fund allocations efficiently across their full urban hierarchy, not just their largest cities.

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Private sector: building bankable, not just compliant, projects

Revenue certainty has been the binding constraint on private participation in Indian SWM. The task for the next decade is converting that diagnosis into a standard practice: DPRs written to UCF's bankability criteria from the outset, tipping-fee and minimum-feedstock clauses negotiated as standard contract terms rather than case by case, and blended structures that combine UCF's 25 per cent central assistance with commercial debt and, increasingly, carbon-linked revenue from methane avoidance. Private operators that build this discipline into their bid preparation, rather than treating financial structuring as a problem for the ULB to solve, can win a disproportionate share of the SPV and concession opportunities the Supreme Court's endorsed model is about to generate at scale.

Investors: underwriting blended and carbon-linked capital

UCF's own design, a 25 per cent central assistance against a minimum of 50 per cent market mobilisation, is effectively an invitation for institutional capital to build a dedicated SWM asset class rather than treating waste infrastructure as a generic municipal bond line item. Two features make this asset class distinctive. First, the underlying revenue streams, tipping fees, RDF offtake, Bio-CNG sales, are now backed by an unusually dense stack of policy support: mandatory RDF ramp-up, the Compressed Biogas Obligation, and DRAP's land-reclamation programme all create demand-side certainty that many emerging-market infrastructure debt lacks. Second, biomining and landfill diversion projects generate a genuine, measurable carbon avoidance case, methane that can otherwise vent from an unscientific dumpsite, which positions well-structured SWM assets to draw voluntary and compliance carbon finance alongside conventional project debt. Investors who build the underwriting capability to price both revenue streams together can have a first-mover advantage as India's SWM asset class matures over this decade.

Technology providers: from dashboards to decisions

The digital investment made across India's SWM system to date, the CPCB portal, GOBARdhan's tracking dashboard, state-level MIS systems, has been overwhelmingly about data collection: registering entities, logging tonnages, verifying compliance. That was the right first investment, but it is not the end state. The next focus of technology investment needs to shift from reporting what happened to shaping what happens next: predictive fleet and route planning that responds to real segregation patterns rather than fixed schedules, asset performance monitoring that flags an underperforming MRF or CBG plant before a number of missed targets accumulates, and landfill analytics that can forecast biomining throughput site by site rather than relying on programme-level averages.

AI and machine learning can significantly improve the management of solid waste contracts by increasing transparency, accountability and coordination. AI-enabled monitoring systems can analyse weighbridge data, GPS records, vehicle movements and tipping transactions to identify anomalies, reduce over-reporting of waste quantities and limit unjustified tipping-fee claims. Predictive analytics can support cluster-based planning by aligning waste generation, transport routes, processing capacity and inter-agency responsibilities across jurisdictions. Machine learning tools can also continuously track contractor performance against complex service-level agreements, flag compliance risks and automate reporting. For private sector participation, AI can provide real-time operational insights, improve revenue forecasting and reduce information asymmetry, creating greater confidence for investors, lenders and public agencies while lowering contract-management costs.

The next phase of India's solid waste management journey may not be defined by another regulatory reform. It may be defined by whether ULBs, states, Technical Support Units, private operators, investors and technology providers can convert a largely settled policy architecture into delivery at the scale of 165 million tonnes of annual waste by 2030. Success can depend less on new rules and more on professionalised institutions, regional infrastructure planning, decision-grade digital systems, and genuinely investable project structures. The cities and states that treat solid waste as a professionally governed infrastructure service, rather than a sanitation activity administered department by department, are the ones likely to set the benchmark for the next decade of urban India.

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Section 4:

Extended Producer Responsibility and the next industrial opportunity



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Extended Producer Responsibility has moved, in the space of roughly three years, from a compliance checkbox to the primary financing mechanism for five separate industrial value chains: plastics, construction and demolition debris, e-waste, textiles and tyres. MoEFCC has itself projected that India's circular economy could represent a USD 2 trillion market opportunity and 10 million jobs by 2050[55]. The gap this section examines is the one between that scale of ambition and the formal recovery infrastructure currently built to capture it.

4.1 EPR on paper vs. EPR in practice: closing India's accountability gap

Plastic packaging is among India's more developed EPR frameworks. Multiple policy revisions have improved the system over time. At the same time, they have highlighted gaps between regulatory objectives and on-ground enforcement.

EPR compliance: registration, targets, and the enforcement shortfall

Every Producer, Importer and Brand Owner (PIBO) must register on the CPCB's EPR portal. Annual EPR targets are then calculated based on the average quantity of plastic packaging introduced into the market during the previous two years. The calculation also includes pre-consumer waste generated. Quantities supplied onwards for further use are deducted from the total. This is commonly expressed as EPR Target = (A + B) minus C[56]. Category-wise recycling targets step up sharply: rigid and compostable packaging (Category I and IV) rises from 50 per cent in FY 2025-26 to 80 per cent by FY 2027-28, while flexible and multi-layered packaging (Category II and III) rises from 30 per cent to 60 per cent over the same window.[57]

Since 1 July 2025, importers without valid EPR registration have had consignments blocked at customs, a meaningfully sharper enforcement lever than the environmental-compensation fines that preceded it

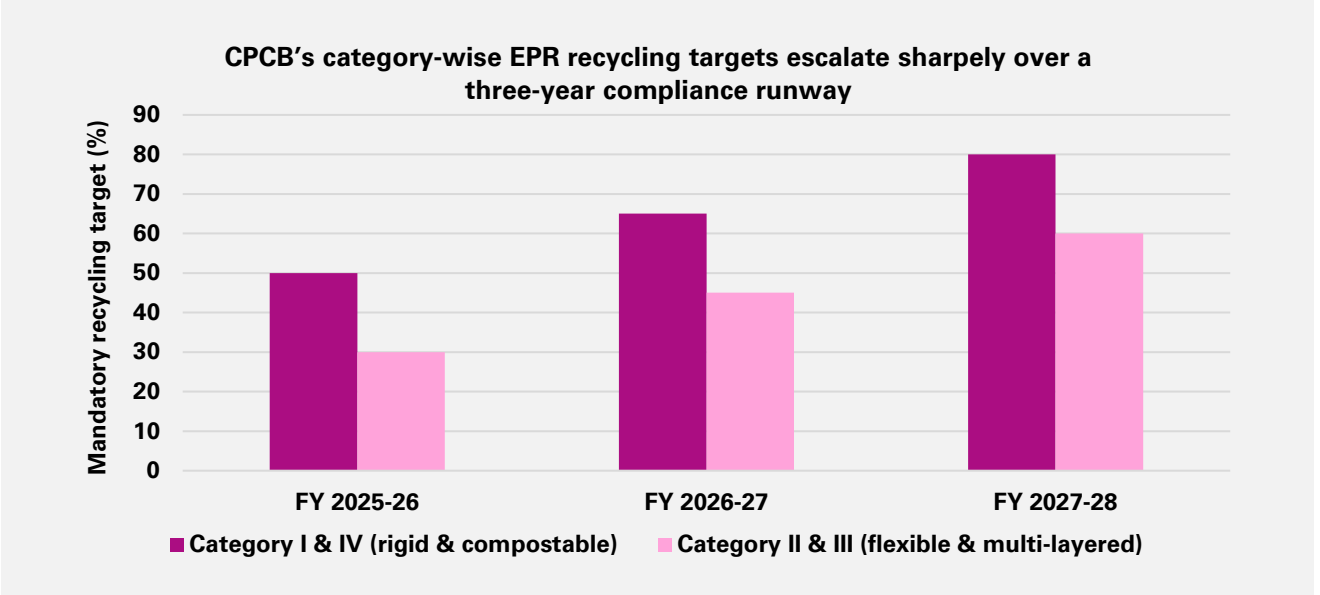


Figure 4.1: CPCB category-wise EPR recycling targets for plastic packaging {{27,28}}

55. Operation Shuddhi, "EPR Compliance in India: A 2026 Guide to Plastic Packaging Rules, Targets & Plastic Credits," 26 June 2026, citing MoEFCC (2025) circular economy market projections.
56. Afleo, "EPR Plastic Waste Registration in India (2026): CPCB Guide for PIBOs," 9 June 2026.
57. GreenSutra, "Plastic EPR Explained 2026: CPCB Targets, Certificates and the PWM Amendment," 14 June 2026, on the Plastic Waste Management (Amendment) Rules, 2026 (notified 31 March 2026).

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Producer accountability: tracing obligations through the value chain

The Plastic Waste Management (Amendment) Rules, 2026 introduced important changes to certificate issuance, audits and the treatment of compliance shortfalls. As a result, many compliance approaches that worked under earlier regulations may no longer be sufficient. The rules also strengthen traceability. Since July 2025, all plastic packaging must carry a QR code, barcode or other unique identifier. This allows each batch of packaging to be linked to its registered producer and tracked through the CPCB portal. For the first time, claims related to recycled content can be verified using digital records rather than relying solely on self-reporting.[58] Central Institute of Petrochemicals Engineering & Technology (CIPET) projects that full compliance with the 2025 target set could cut macroplastic leakage by 35 per cent by 2027 and reduce GHG emissions by 4.5 million tonnes of CO2-equivalent annually, a useful reminder that EPR enforcement is climate policy as much as it is waste policy.

Credits and certificates: building a credible EPR marketplace

Registered recyclers and processors receive EPR certificates based on the quantity of waste they have processed and verified. Producers, Importers and Brand Owners (PIBOs) purchase these certificates to meet their EPR obligations. Unlike the e-waste and battery sectors, certificate prices in the plastic waste market are not regulated. Prices are determined by market demand and supply. When a PIBO falls short, environmental compensation escalates from INR5,000 per tonne in the first year of shortfall to INR10,000 in the second and INR20,000 in the third, with unfulfilled targets carried forward for up to three years and partial compensation refunds, 75 per cent, 60 per cent and 40 per cent respectively, available if the shortfall is subsequently cleared.

This escalating structure is a deliberate design choice: it makes chronic non-compliance progressively more expensive than simply buying certificates, which is precisely the price signal a credible EPR marketplace needs.

4.2 Building the value chain: plastics, C&D, e-waste, textile, and tyre as India's next industrial opportunity

A comparison of India's five priority EPR value chains reveal significant differences in their level of maturity and formalisation. These differences highlight where future policy attention and investment are needed.

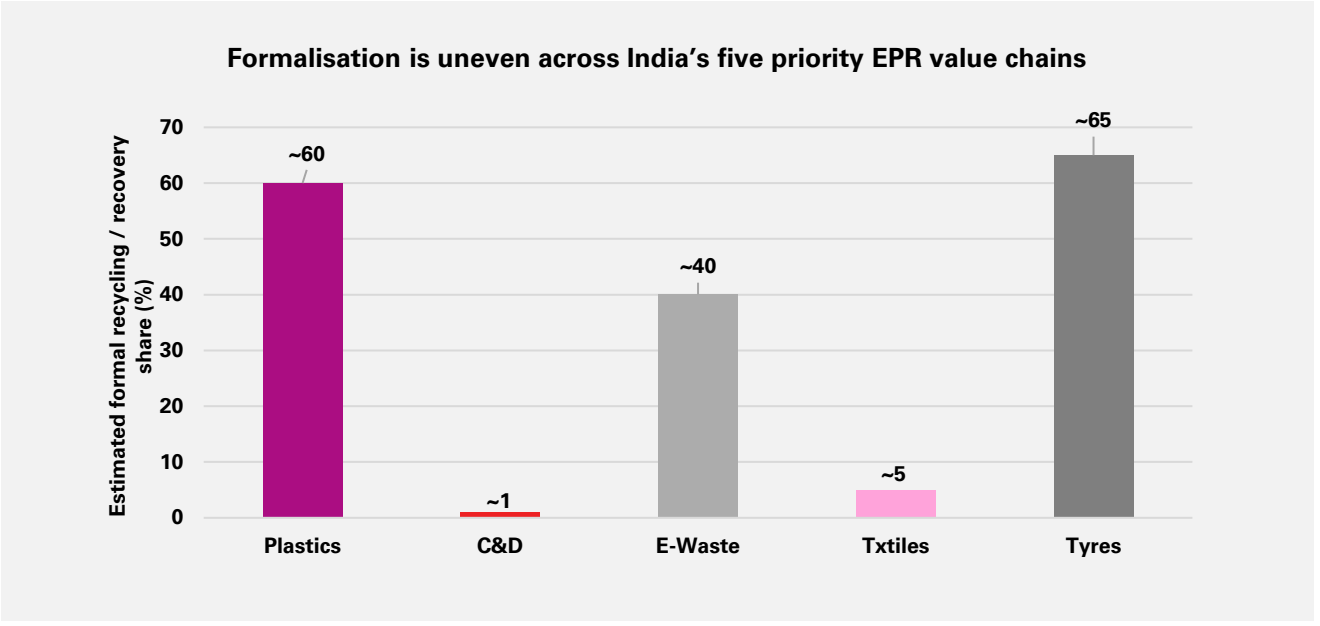


Figure 4.2: Illustrative formal recycling/recovery share by waste stream {{54.56}} (estimates vary by source; see text)

58. Recircle, "India's Plastic Waste Rules 2025: What You Must Know," 4 November 2025, citing Central Institute of Plastics Engineering and Technology (CIPET) projections.

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Plastics: collection, sorting, and recycling infrastructure gaps

India's plastic recycling market processed an estimated 3.8 million tonnes in 2023 and is projected to reach 5.5 million tonnes by 2030, while the broader plastic waste management market is expected to grow from roughly USD1.73 billion in 2025 to USD2.33 billion by 2030[59].

Per capita plastic use is approaching 11 kilograms annually, and over 5.8 million tonnes of plastic waste are still incinerated each year rather than recycled, underlining that market growth and infrastructure adequacy are not the same thing.

C&D waste: processing capacity vs. urban construction volumes

Construction and demolition waste remains the least understood of India's priority waste streams. Annual generation estimates range from 150 million to 500 million tonnes, reflecting major gaps in measurement and reporting. Without reliable data, planning, regulation and investment decisions remain difficult[60]. What is not contested is the recovery rate: the Centre for Science and Environment estimates barely 1 per cent of C&D waste is recycled nationally. Of 53 cities expected to have C&D recycling facilities by 2020, only 13 delivered them, with a combined capacity of about 6,500 tonnes per day against generation running into the tens of thousands of tonnes[61].

The Construction and Demolition Waste Management Rules, effective 1 April 2026, now bring an EPR-style obligation to construction: projects above 20,000 square metres of built-up area are classified as producers and face recycling targets rising from 25 per cent to 100 per cent by FY 2028-29, alongside a mandatory 5 to 25 per cent recycled-content requirement in new construction.

E-waste: formal recovery infrastructure and critical mineral recovery

India is the world's third-largest producer of e-waste, generating an estimated 1.75 million tonnes in FY 2023-24, up 73 per cent in five years, alongside roughly 60 kilotonnes of spent lithium-ion batteries annually[62] [63]. Estimates of the formal recycling share diverge sharply depending on scope and methodology: CPCB data suggests that formal e-waste recycling has expanded significantly, increasing by around 240 per cent since 2019 and exceeding 40 per cent of reported volumes. However, other assessments using higher estimates of total e-waste generation place formal recovery closer to 10 per cent. This highlights a wider challenge around data consistency and measurement. Despite recent progress, collection and initial dismantling activities continue to take place in the informal sector, which remains the dominant participant in the e-waste value chain[64].

The economic opportunity is substantial. India's e-waste stream is estimated to contain materials worth about INR51,000 crore, with nearly INR30,600 crore considered recoverable. However, a large share of this value remains unrealised. The informal sector currently captures more value than the formal sector, while significant economic potential is lost due to inefficient collection, recovery and processing practices. Strengthening formal recycling systems could therefore deliver both environmental benefits and greater resource value recovery. The government's response has been a INR1,500 crore incentive scheme for critical mineral recycling, approved by Cabinet on 3 September 2025 under the National Critical Mineral Mission, offering recyclers up to INR50 crore (large) or INR25 crore (smaller) to invest in end-to-end hydrometallurgical recovery of lithium, cobalt and nickel, rather than exporting semi-processed black mass as most of the sector does today.

59. BlueWeave Consulting, "India Plastic Recycling Market Share & Growth Analysis (2024-2030)."
60. Citizen consumer and civic Action Group (CAG), "India's construction boom and the waste challenge: Policies, gaps, and data," 13 January 2026, citing Ministry of Housing and Urban Affairs and Centre for Science and Environment estimates.
61. Site Setu, "C&D Waste Management Rules 2025 India: EPR, Recycling Targets, IS 383, CPCB Compliance Guide," 21 June 2026, on G.S.R. 219(E).
62. Deccan Chronicle, "India Pushes for E-Waste Recycling to Recover Critical Minerals," 25 October 2025, citing Ministry of Mines data and the Rs 1,500 crore Incentive Scheme approved 3 September 2025.
63. Context (Thomson Reuters Foundation), "In-Data: E-waste recycling could provide India's critical minerals," 21 October 2025, citing United Nations and Central Pollution Control Board figures.
64. Down To Earth, "India's E-Waste Goldmine: Rs 51,000 Crore in Materials, But Most Value Slips Through the Recycling System," 4 May 2026.

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Textile recycling: an emerging category with near-zero formal infrastructure

The Ministry of Textiles estimates that India generates about 70.7 lakh tonnes of textile waste each year. Around 42 per cent comes from manufacturing processes, while 58 per cent is generated after consumer use. This waste stream represents a significant circular-economy opportunity. Formal recycling could create a market worth USD3.5 billion and support approximately one lakh green jobs by 2030[65].

Infrastructure today is concentrated in a small number of hubs: Panipat alone handles 3,500 to 5,250 tonnes per day as India's principal textile-to-textile recycling hub, while Delhi's informal Katran Market at Mongolpuri sorts over 10 tonnes per day of cutting waste by colour before it reaches formal recyclers, illustrating how thoroughly the informal sector still underwrites this category[66].

Tyre recycling: pyrolysis, crumb rubber, and the road ahead for end-of-life tyres

Tyre EPR is among the most mature EPR frameworks in India. Producer obligations increased from 35 per cent in 2022-23 to 100 per cent by FY 2024-25. Producers meet these obligations by purchasing EPR certificates from CPCB-registered recyclers. These recyclers recover value from end-of-life tyres through products such as crumb rubber, reclaimed rubber and pyrolysis oil[67]. The underlying generation data, however, is unusually inconsistent across sources: industry estimates for 2025-26 range from 1.5 to 1.8 million tonnes of annual waste tyre arisings, while a separate CPCB-linked trajectory shows generation growing from about 220,000 tonnes in 2015 to a projected 395,000 tonnes in 2025 and 600,000 tonnes by 2030[68].

That order-of-magnitude gap between estimates is itself a data-governance signal worth flagging: EPR credit markets can only price risk correctly once generation baselines are standardised, and tyres are not the only stream, C&D waste shows the same pattern, where this remains unresolved.

Value chain	Annual generation (India)	Formalisation signal
Plastics	~3.8 to 5.5 million tonnes recycled (2023 to 2030E)	Market-driven certificate pricing; QR-code traceability from Jul 2025
C&D waste	150 to 500 million tonnes (wide data gap)	~1% recycled nationally; Delhi an outlier at ~80%
E-waste	1.75 million tonnes (+60 kt spent LIBs)	Formal share estimated 10% to 40%+ depending on methodology
Textiles	70.73 lakh tonnes (42% pre / 58% post-consumer)	Formal capacity concentrated in Panipat hub
Tyres	0.4 to 1.8 million tonnes (estimates vary)	100% producer EPR obligation live since FY 2024-25

Table 4.1: India's five priority EPR value chains, generation and formalisation at a glance (66,68)

4.3 The secondary materials market India needs to build

EPR creates the legal obligation to recycle; it does not, by itself, create a liquid, trusted market for the recycled material that results. That second, missing layer is the secondary materials market.

Price volatility and quality standards in recycled material markets

Recycled plastic markets in India remain largely market-driven. Prices for both recycled materials and EPR certificates fluctuate with virgin resin prices, feedstock availability and collection volumes. This uncertainty can discourage long-term investment in recycling infrastructure. Quality standards are beginning to strengthen the market. Recycled plastics used in food-contact applications must now comply with IS 14534:2023 and carry FSSAI

65. India Brand Equity Foundation (IBEF), "India's textile recycling market is projected to reach US$ 3.5 billion by 2030," 11 March 2026, citing Ministry of Textiles, "Mapping of Textile Waste Value Chain in India."
66. ESG News Earth, "India Maps Waste to Rule Textile Export Market."
67. My Waste Solution, "Tyre Recycling Business in India: Cost, Setup & Profit (2026)," 12 May 2026.
68. A Series, "Waste Tyre EPR in India: Turning a Challenge into Circular Opportunities," citing CPCB monitoring data.

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certification. This provides a clear and verifiable standard for buyers, helping shift recycled content from a sustainability claim to a certifiable product specification.

Demand-side pull: procurement mandates for recycled content

Recycled-content mandates that took effect in 2025-26 for plastic packaging, and the 5 to 25 per cent recycled-content requirement now built into the C&D Waste Management Rules for new construction, represent India's first systematic attempt to create demand-side pull for secondary materials rather than relying on supply-side collection mandates alone.

India's circular economy overall is projected to generate an additional USD218 billion in annual value by 2030, rising to USD624 billion by 2050, a scale of opportunity that can only be realised if procurement, public and private, treats recycled content as a specification requirement rather than a sustainability add-on[69].

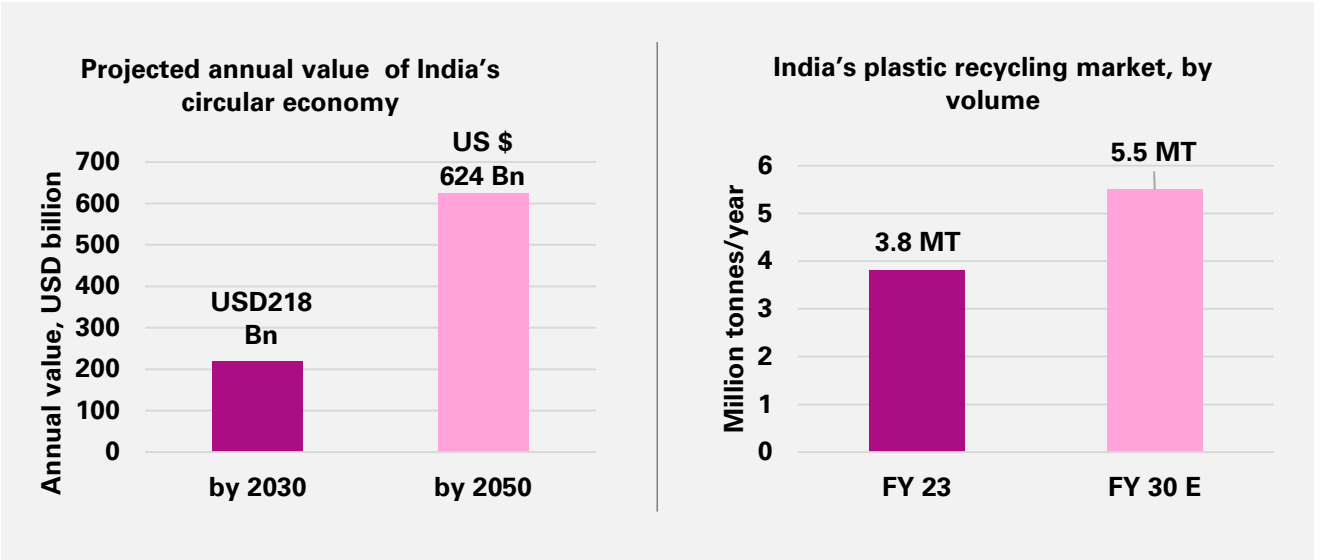


Figure 4.3: India's projected circular economy value and plastic recycling market growth {{66,67}}

Building trust: certification and traceability for secondary materials

Traceability infrastructure, the QR-coded packaging mandate, the CPCB digital portal, and IS-standard certification for recycled content, is converging toward the same goal: making a recycled-content claim independently verifiable rather than self-reported. India's plastics circularity sector alone has already attracted INR 65,388 million in investment across 199 deals between 2018 and 2023, spread across 4,953 registered and 823 non-registered manufacturing and recycling units in 30 states and Union Territories, evidence that capital is prepared to back this market once the traceability and standards infrastructure gives it a reason to trust the underlying material claims[70].



69. India Brand Equity Foundation (IBEF), "Circular Economy for Sustainable Development in India," citing NITI Aayog-linked estimates of India's circular economy value potential.

70. UJA, "Plastic Industry in India" market report, 14 June 2025, citing Central Pollution Control Board registration data and investment tracking (2018-2023).

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4.4 Formal meets informal: circular business models that include the informal sector

No India-specific circular economy strategy is complete without an explicit position on the informal sector, because in India, unlike most of the markets EPR regulation was originally designed for, the informal sector is not a gap in the system. It is the system's largest existing node of material recovery.

Waste picker integration: livelihoods, recognition, and formalisation pathways

India's informal waste recycling economy, waste pickers, kabadiwalas, aggregators and intermediate dealers, is estimated at around 4 million people[71]. In Delhi alone, informal waste pickers recover an estimated 20 to 25 per cent of the roughly 10,000 metric tonnes of waste the city generates daily, work that is currently unrecognised, largely unprotected, and increasingly threatened as private formal-sector players expand into areas informal collectors have long served[72].

Waste-pickers and informal enterprises play a critical role in achieving recycling outcomes, but often lack access to social protection, occupational safety, finance, formal markets and stable livelihoods. A truly circular economy must therefore be inclusive by design. Integrating informal workers into formal systems, strengthening their capacities and improving working conditions are not only social imperatives, but also essential for building efficient, resilient and scalable resource-recovery systems.

The government's NAMASTE programme now includes a dedicated Waste Pickers component for FY 2024-25 and FY2025-26, covering identification and ID-card issuance, occupational safety training and PPE, and linkage to social security schemes, the first systematic national attempt to formally enumerate this workforce[73].

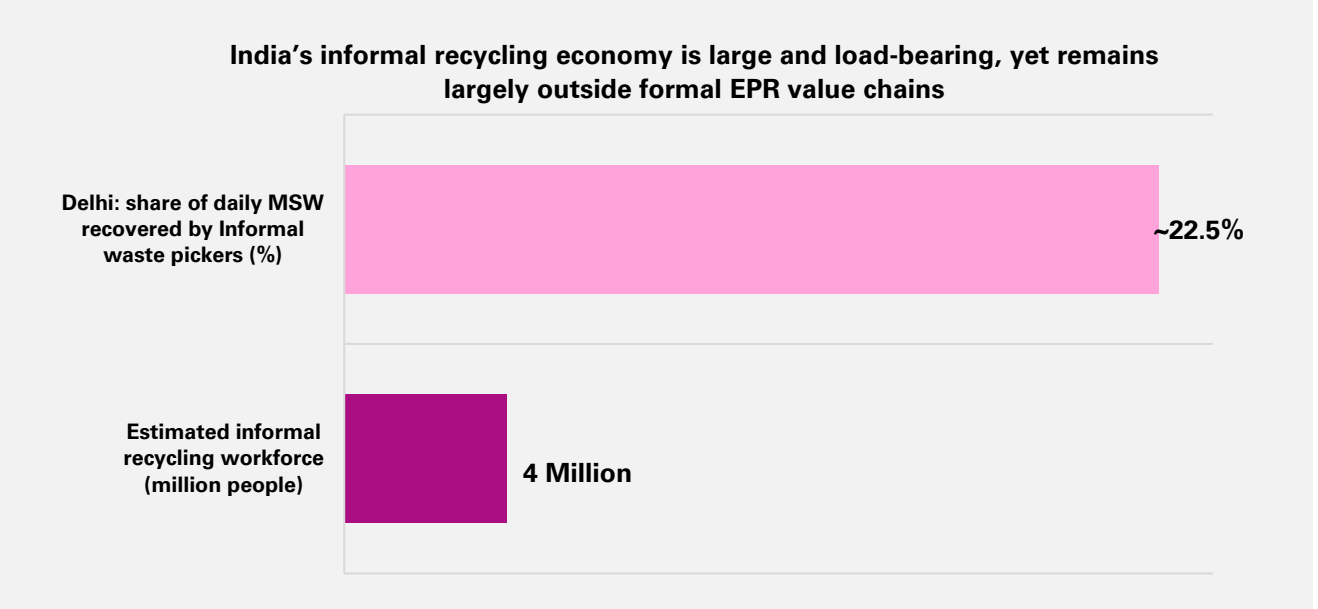


Figure 4.4: The scale of India's informal recycling economy {{68,69}}

71. International Alliance of Waste Pickers, "Organising and Integrating the Informal Recycling Sector in the Solid Waste Management Sector - India," 7 January 2026.
72. British Safety Council India, "India's waste pickers: indispensable but invisible," 7 February 2025.
73. Ministry of Social Justice and Empowerment, Government of India, "Scheme Guidelines For inclusion of Waste Pickers Component under NAMASTE," FY 2024-25 to FY 2025-26.

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UNDP's Project Utthaan offers a working template for what bridging looks like in practice: operational across 35 cities, it has processed over 83,000 metric tonnes of plastic to date and directly reached 5,000 Safai Sathis in its first phase. Building on this momentum, a second phase supported through corporate partnerships is extending outreach to 3,000 additional workers across Delhi and Mumbai through four dedicated Swachhta Kendras serving 75,000 households[74].

Pune's SWaCH cooperative remains the reference model nationally for integration rather than displacement, built on sustained political will, administrative support and active waste-picker participation in decision-making rather than a top-down absorption into municipal payrolls[75].

Investable circular models: from CSR pilots to scalable enterprises

Experience across waste management value chains shows that the informal sector is often the foundation of collection and material recovery systems. Building parallel collection networks from scratch is typically costly, slow and difficult to scale. In contrast, models that strengthen and organise existing informal networks can achieve faster coverage, higher recovery rates and better economics.

This creates a clear investment opportunity. Infrastructure such as Material Recovery Facilities (MRFs), aggregation centres, buy-back centres and EPR-backed collection systems perform best when they are designed to work with existing waste-picker networks, cooperatives, self-help groups and informal enterprises. The objective is not to replace the informal sector, but to improve its productivity, safety and market access.

A similar lesson applies to corporate participation. For many producers and brand owners, partnering with and expanding trusted collection networks can be more effective than creating independent collection systems. The long-term opportunity lies in converting such partnerships into structured commercial arrangements with clear performance targets, pricing mechanisms, offtake commitments and risk-sharing frameworks. This approach can simultaneously improve recycling outcomes, strengthen livelihoods and create scalable, bankable circular economy businesses.

Corporate philanthropic capital can play a catalytic role when structured as guarantees, first-loss facilities, viability gap support or outcome-based funding rather than as standalone grants. Such instruments can reduce project risk, improve bankability and attract commercial investors who may otherwise remain hesitant. This approach can significantly increase the impact achieved per rupee of philanthropic funding while mobilising additional private capital through blended finance structures. The success of this model, however, depends on rigorous project structuring. Clear revenue streams, risk allocation, performance metrics and governance arrangements are essential to convert promising interventions into investable opportunities capable of attracting long-term financing at scale.

4.5 From waste targets to material markets

India has created a strong regulatory framework for material recovery. EPR obligations now cover plastics, e-waste, tyres and construction and demolition waste. However, recovery alone is not enough. Markets are needed to absorb and use the recovered materials. Many supporting elements are still developing, including quality standards, certification systems, procurement mechanisms, trading platforms and reliable buyers. Without these, recovered plastic, recycled aggregates, critical minerals and recycled fibres cannot compete effectively with virgin materials.

India's next challenge is therefore not creating recovery obligations, but building markets for recovered resources. In this sense, EPR is evolving beyond environmental compliance into a broader strategy for resource security, industrial development and circular economic growth.

74. United Nations Development Programme, "UNDP releases first-of-its-kind assessment of the socio-economic status of waste pickers in India," Project Utthaan.

75. Down To Earth, "Integration or formalisation of informal waste pickers: Beyond the binary," 1 April 2025, on the SWaCH cooperative model in Pune.

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Building the market layer: exchanges, standards and certification

India's recycling system is still focused on recovery rather than market creation. Waste is collected, processed and converted into secondary materials, but markets for trading these materials remain underdeveloped. More mature circular economies support recycling with common standards, quality certification, procurement frameworks and established buyer networks. India has made progress in some areas, such as standards for food-grade recycled plastics and organised markets for tyre-derived products. However, many recovered materials still trade through project-specific or bilateral arrangements rather than transparent market mechanisms at scale. The next priority is to build confidence in recovered materials. This require quality standards, certification systems and procurement platforms for products such as recycled aggregates, textile fibres and recovered metals. Stronger markets help convert waste recovery from a compliance activity into a sustainable industrial ecosystem.

Traceability becomes the next infrastructure layer

Investment discussions in India's circular economy focus on physical infrastructure such as recycling plants, MRFs and processing facilities. However, the next major opportunity may lie in building trust infrastructure. The QR code and unique-identifier requirements for plastic packaging are an important first step. They make it possible to track materials and verify their origin. Extending similar systems to e-waste, textiles and construction materials can improve transparency across value chains. This allows buyers to independently verify recycled content, material quality and chain of custody instead of relying on supplier declarations. Stronger traceability can reduce compliance and sourcing risks, build market confidence and support the development of more transparent and efficient secondary-material markets.



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Section 5:

Priority Interventions for stakeholders - Building India's resource-secure and circular future



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Water, wastewater, solid waste and secondary materials should be planned as connected resource systems rather than separate sectors. The objective is not more infrastructure, but better resource productivity.

5.1 A new operating principle: Adopt resource productivity as a national performance metric

India's next challenge is not building more infrastructure. It is delivering more value from the infrastructure already created.

Move beyond measuring:

- Assets created (kilometres of pipeline laid, treatment capacity installed, tonnes of waste collected, project expenditure achieved).

Start measuring:

Water secured and protected

Water reused and recycled

Energy recovered

Materials recovered and returned to productive use

Emissions avoided

Land reclaimed and restored

Service quality, reliability and affordability.

5.2 Water security: Create markets for treated wastewater

Major urban centre should develop:

- Wastewater reuse targets
- Industrial off-take agreements
- Reuse pricing frameworks
- Recycled water distribution systems.

The critical shift

Stop planning treatment plants

Start planning reuse economies.

5.3 Scaling waste as a resource: Scale regional waste economies instead of city-by-city infrastructure

Many smaller cities might never generate enough waste to support standalone facilities economically. Develop:

Multi-city waste clusters

Shared processing infrastructure

Regional biomethanation hubs

Regional MRF networks

State-level waste logistics planning.

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5.4 Create city-level circular economy cells

CBG pricing is improving, carbon markets are emerging locally as well, GOBARdhan is expanding, and resource recovery projects are becoming commercial businesses. Strong technical support is now as important as infrastructure funding.

Cities need access to:

- Market intelligence
- Commercial structuring expertise
- PPP support
- Transaction advisory support.

5.5 Develop a national secondary materials exchange

India has built recovery obligations. It has not yet built commodity markets. The long-term goal should be making secondary materials as easy to procure as virgin materials. The exchange could be built with following focus areas:

- National quality standards
- Material certification systems
- Procurement mandates
- Trading and price-discovery platforms.

5.6 Use philanthropic capital as risk capital

Corporate philanthropy should increasingly provide:

- First-loss support
- Guarantees
- Project preparation facilities
- Outcome-based payments
- Viability-gap support.

This can crowd in significantly larger pools of commercial capital.

5.7 Build decision infrastructure, not dashboard infrastructure

Success should be measured by operational decisions improved, not dashboards deployed. The next generation of digital investments should focus on:

- Predictive maintenance
- Leak detection
- Demand forecasting
- Contract monitoring
- Route optimisation
- Resource-flow analytics.

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Over the past decade, India has demonstrated that it can build environmental infrastructure at scale. The next decade is expected to test whether it can operate that infrastructure as an integrated resource system. Water security, circularity and waste management are converging around a common challenge: managing finite resources more productively. The countries that succeed might not necessarily be those that build the most assets, but those that create effective institutions, trusted markets and efficient resource cycles.



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